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Nakaya

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(54) **FLOW CONTROL VALVE, DAMPER AND STEERING DEVICE**

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B62D 7/22 (2006.01)
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CPC ***B62D 3/12*** (2013.01); ***B62D 7/228*** (2013.01); ***F16F 9/512*** (2013.01); ***F16F 15/023*** (2013.01); ***F16K 17/30*** (2013.01)

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CPC B62D 3/12; B62D 7/228; F16F 9/512; F16F 15/023; F16K 17/30
See application file for complete search history.

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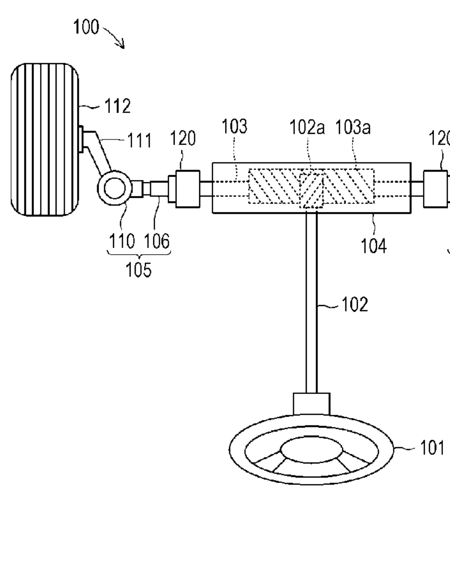
(57) **ABSTRACT**

Provide is a steering device capable of absorbing great impact force, a damper applicable to the steering device, and a flow control valve applicable to the damper. The steering device **100** includes a damper **120** between a rack bar **103** and a rack end **106**. In the damper **120**, an inner chamber **121** is formed at an outer peripheral portion of a socket main body **107**, and an integral displacement body **130** is slidably fitted onto the outer peripheral portion. The integral displacement body **130** is, at an inner peripheral portion thereof, formed with a circular ring-shaped flow control valve **140**. The flow control valve **140** is provided with a first flow control valve **150**. The first flow control valve **150** includes a second flow body **156** that approaches or separates from a first flow body **153**. In the second flow body **156**, a second flow hole **157** is formed at a position shifted from a first flow hole **154** formed at the first flow body **153**, and a second hole diameter restriction portion **158** is formed so as to close the first flow hole **154**.

15 Claims, 14 Drawing Sheets



(51) **Int. Cl.**

F16F 9/512	(2006.01)
F16F 15/023	(2006.01)
F16K 17/30	(2006.01)

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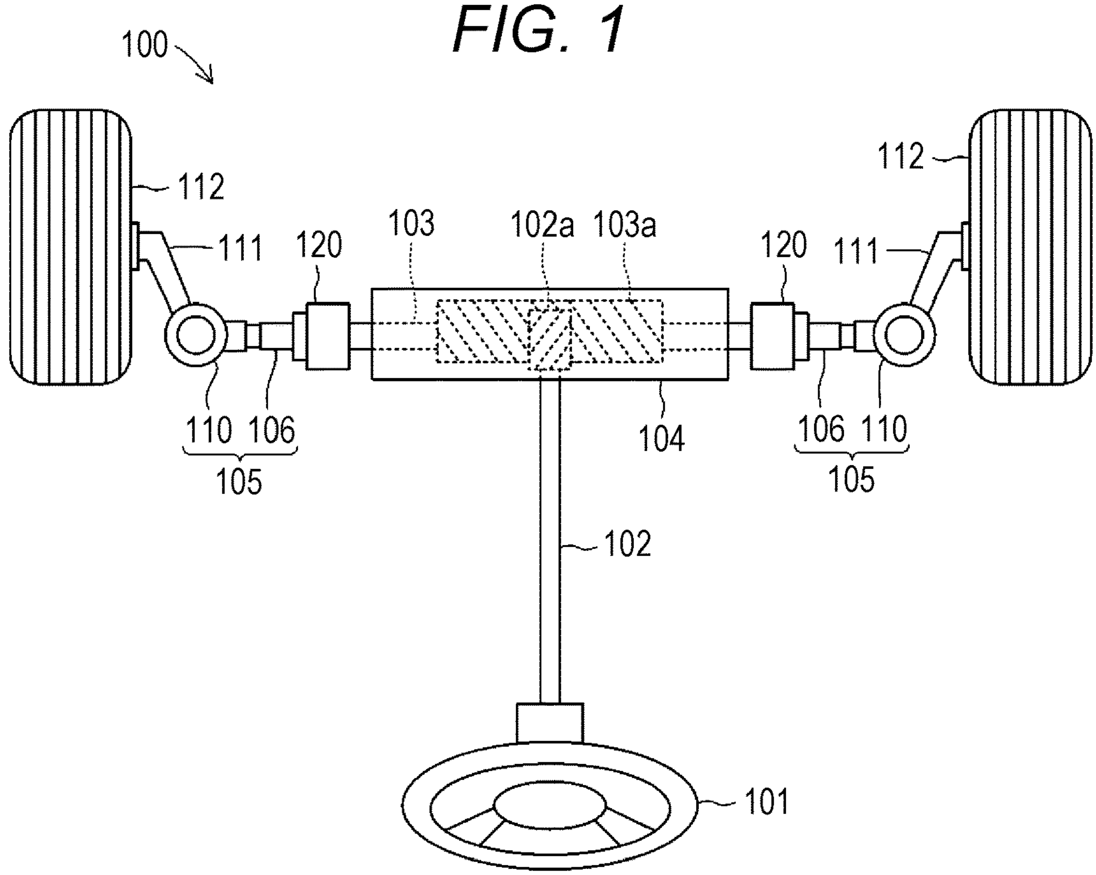
FIG. 1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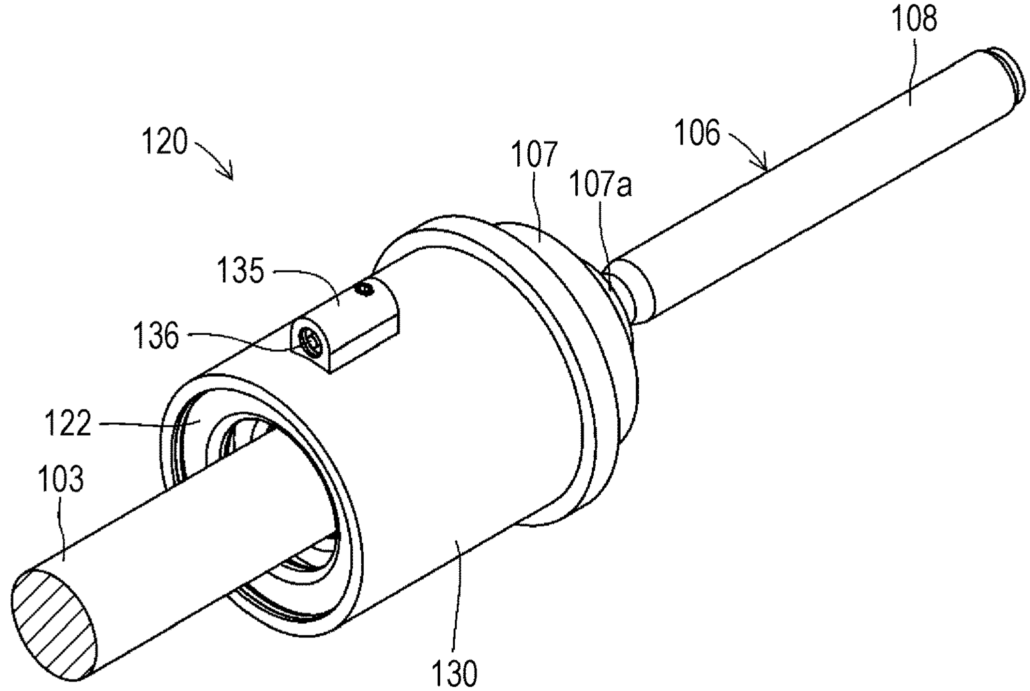
120
103
102a
103a
120
111
112
110 106
105
104
106 110
105
102
101

FIG. 2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120
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107a
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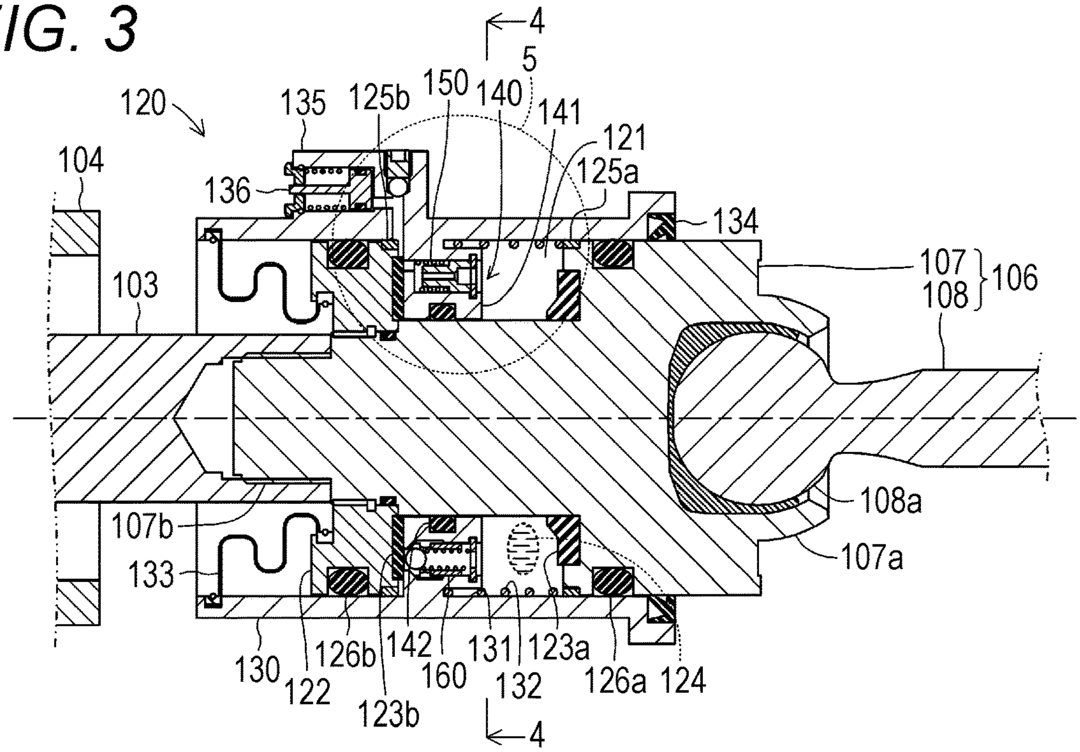
122
103
130

FIG. 3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125b
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125a
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108a
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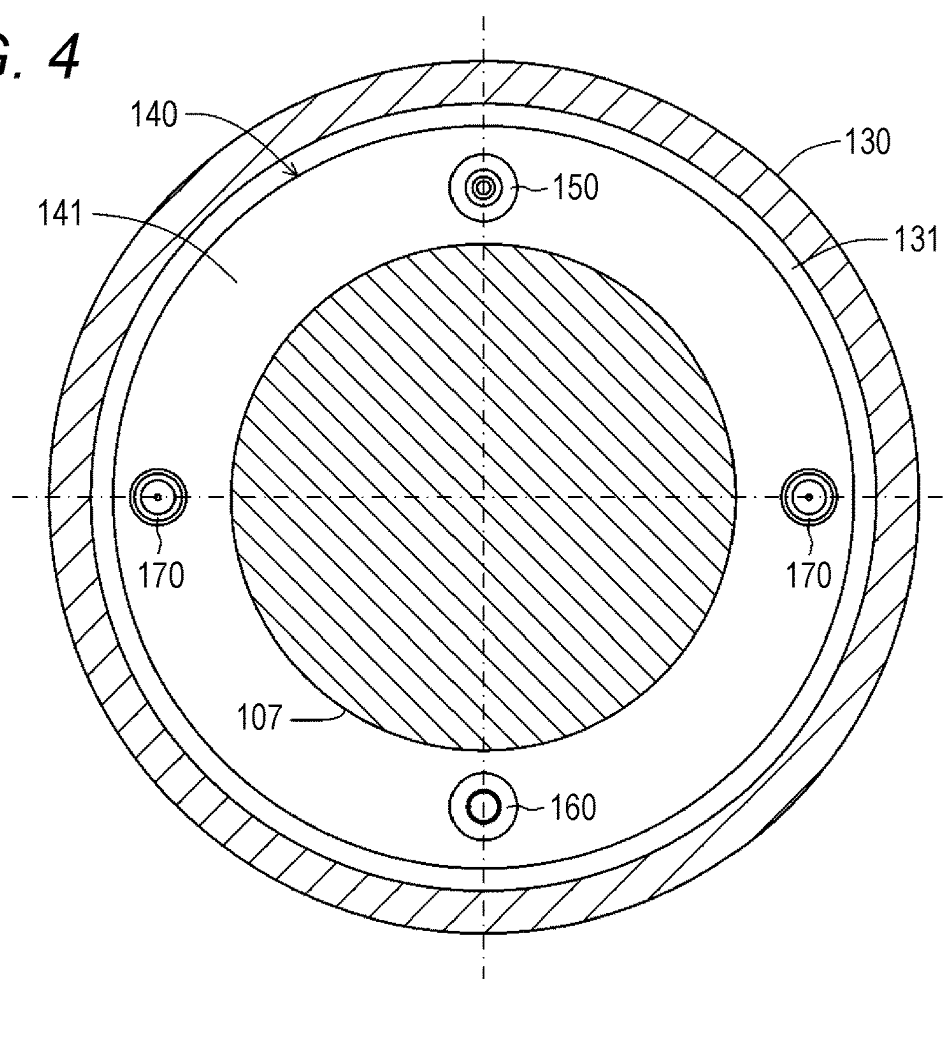
107a
133
130
122
126b
123b
142
160
131
132
123a
126a
124
4

FIG. 4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130
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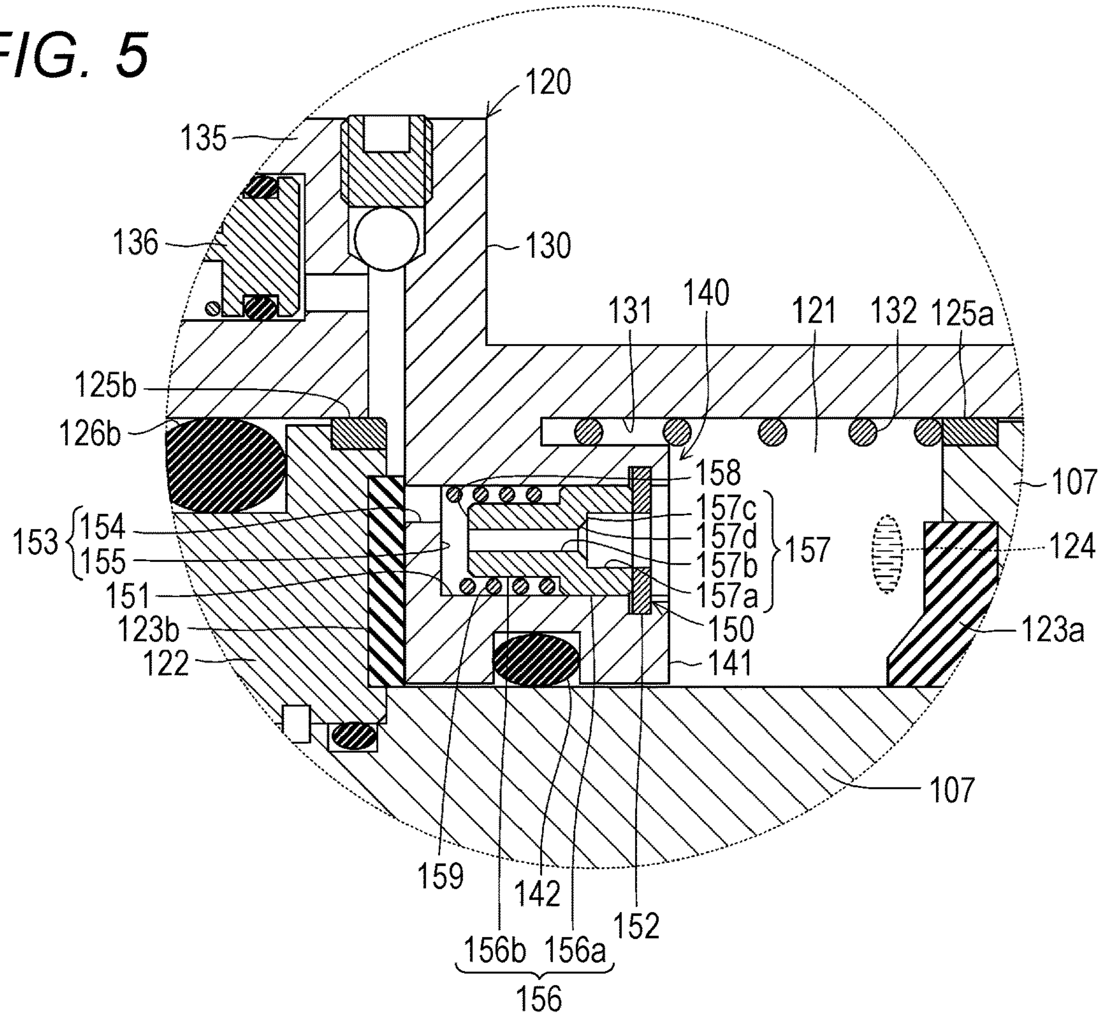
141
131
170
170
107
160

FIG. 5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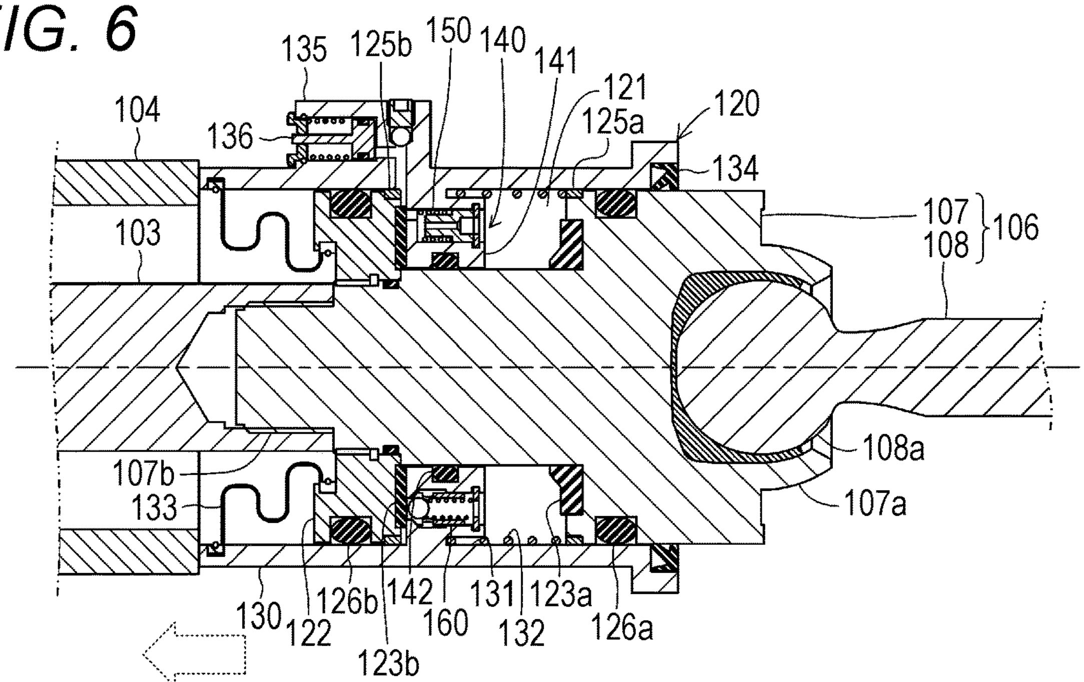
131
140
121
132
125a
125b
126b
107
158
157c
153
154
157d
157
155
157b
124
151
157a
123b
150
123a
122
141
107
159
142
152
156b
156a
156

FIG. 6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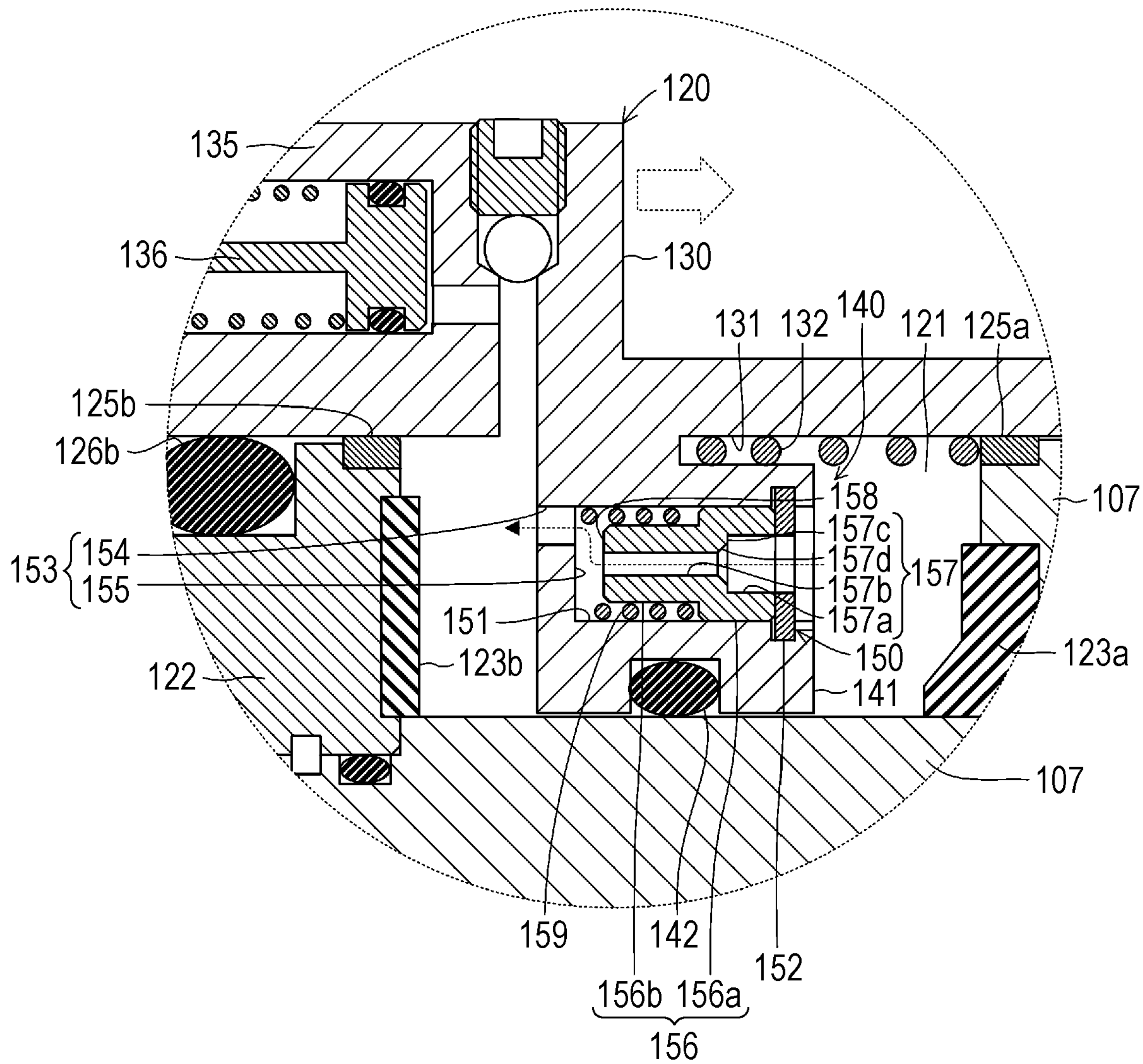
123a
126a
123b

FIG. 7



120
135
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132
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125a
125b
126b
158
157c
157d
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157b
157a
153
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123b
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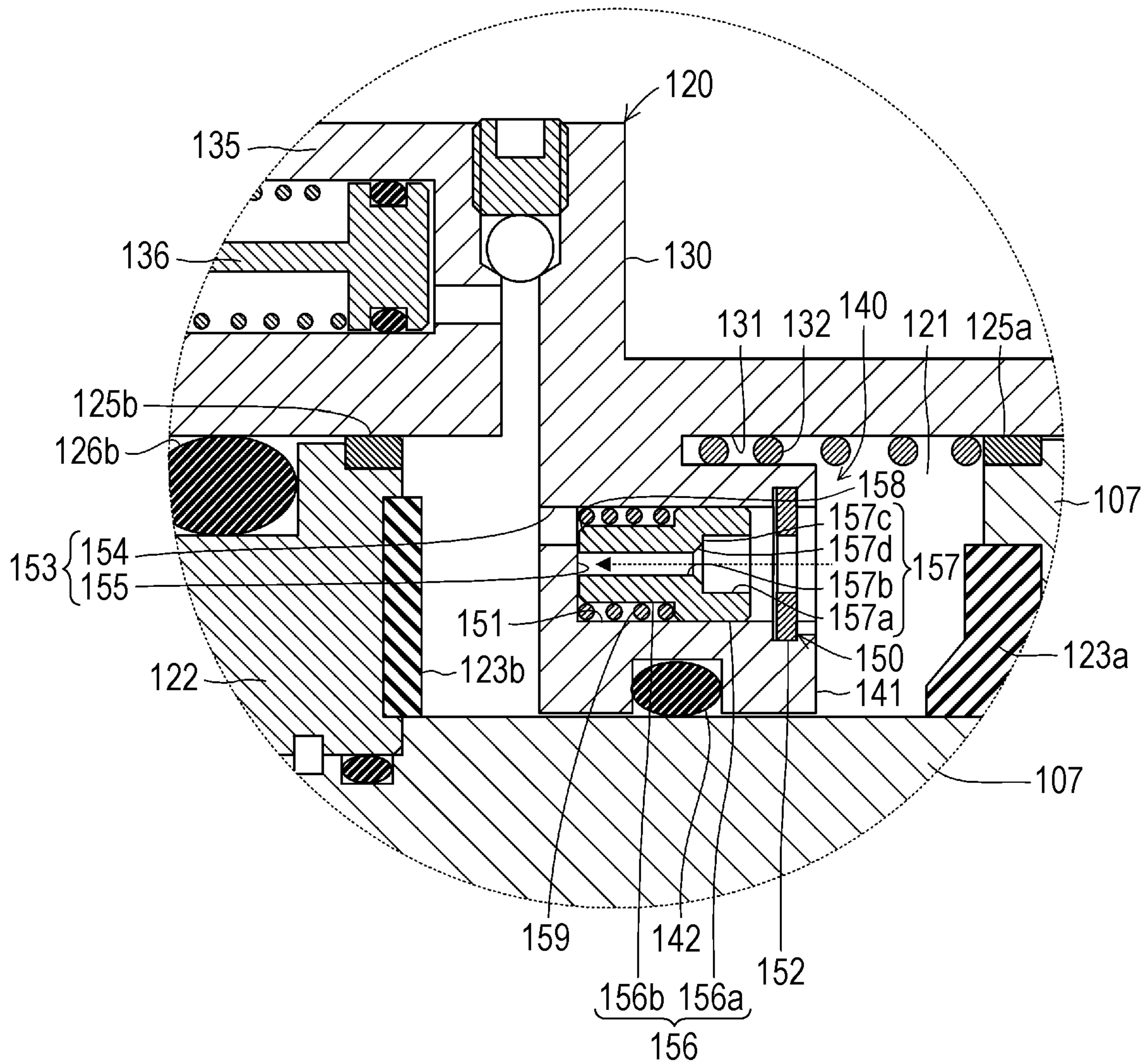
107
107
159
142
152
156b
156a
156

FIG. 8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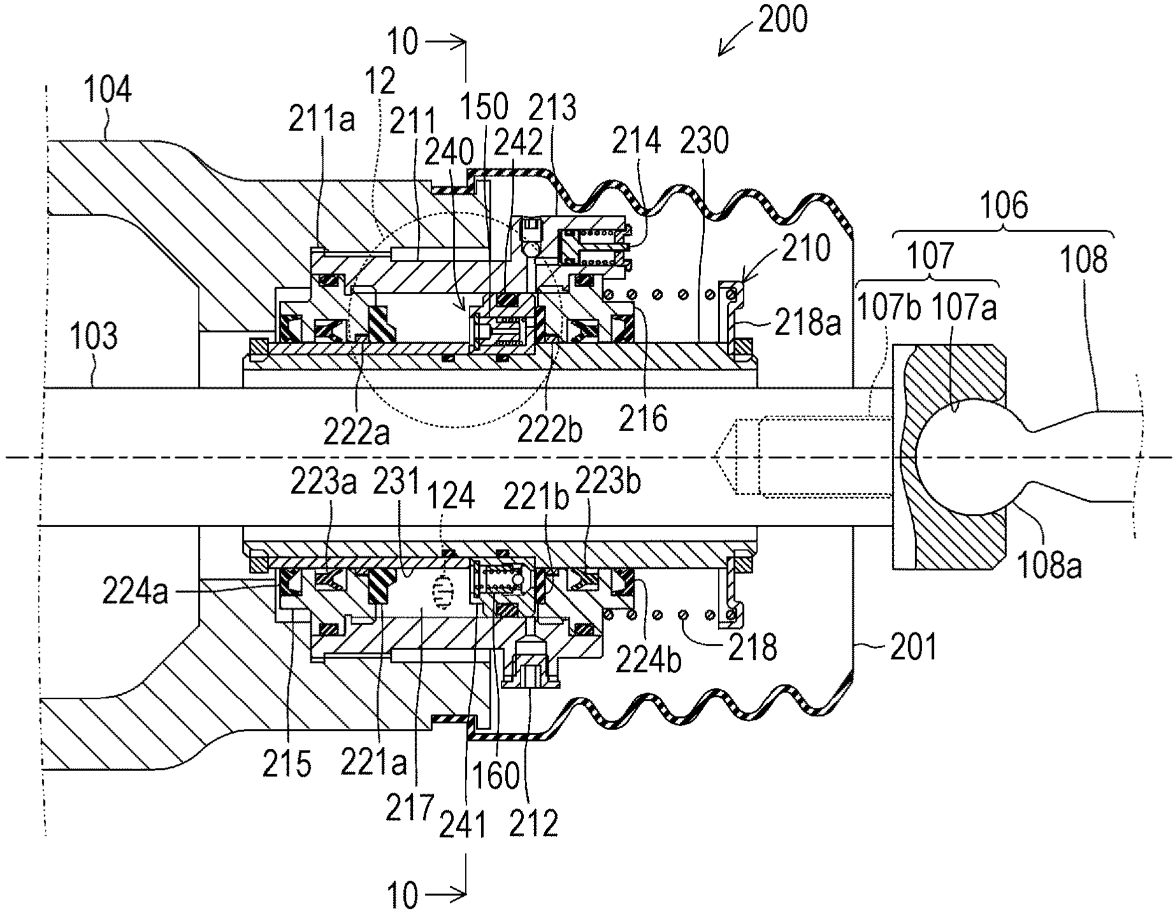
156b
156a
156

FIG. 9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211a
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222a
222b
216
223a
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201
224b
215
221a
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241
212
10

FIG. 10

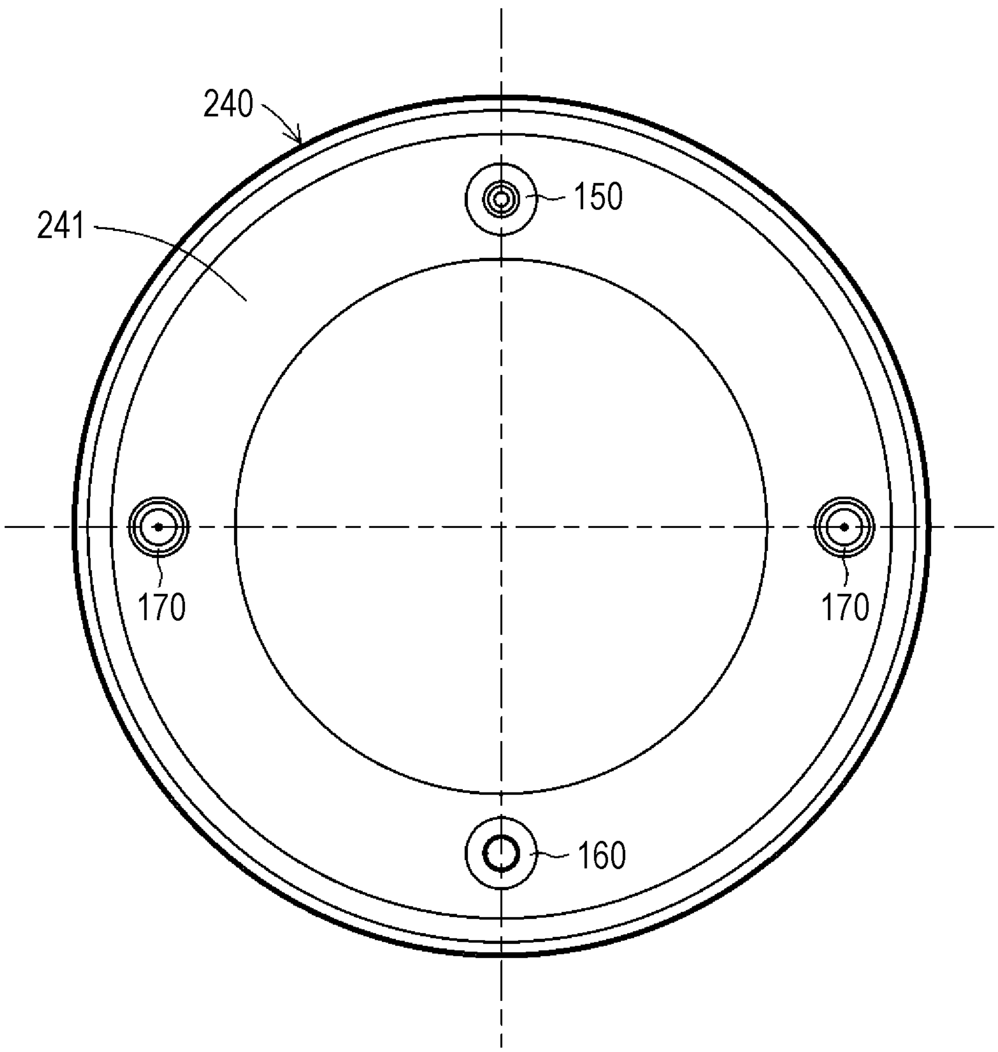
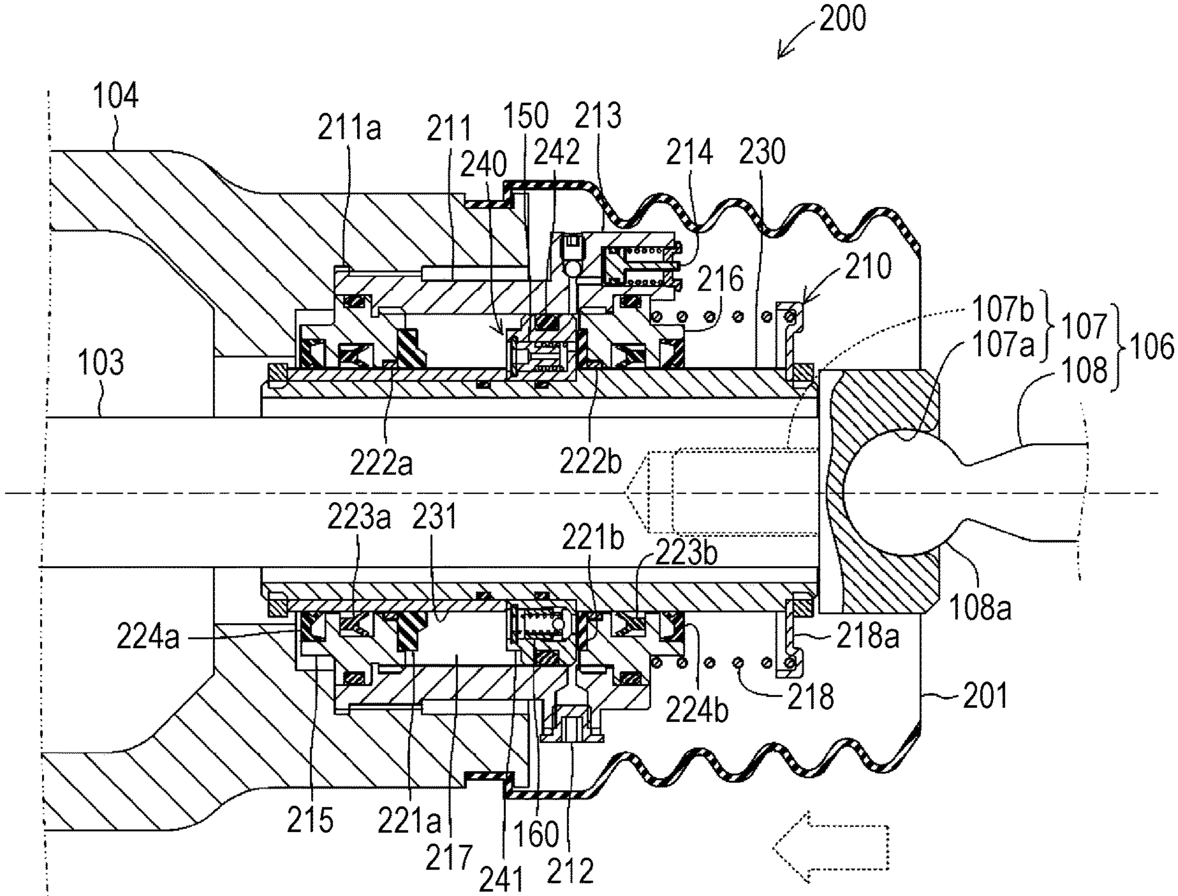


FIG. 11



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222b
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221a
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241
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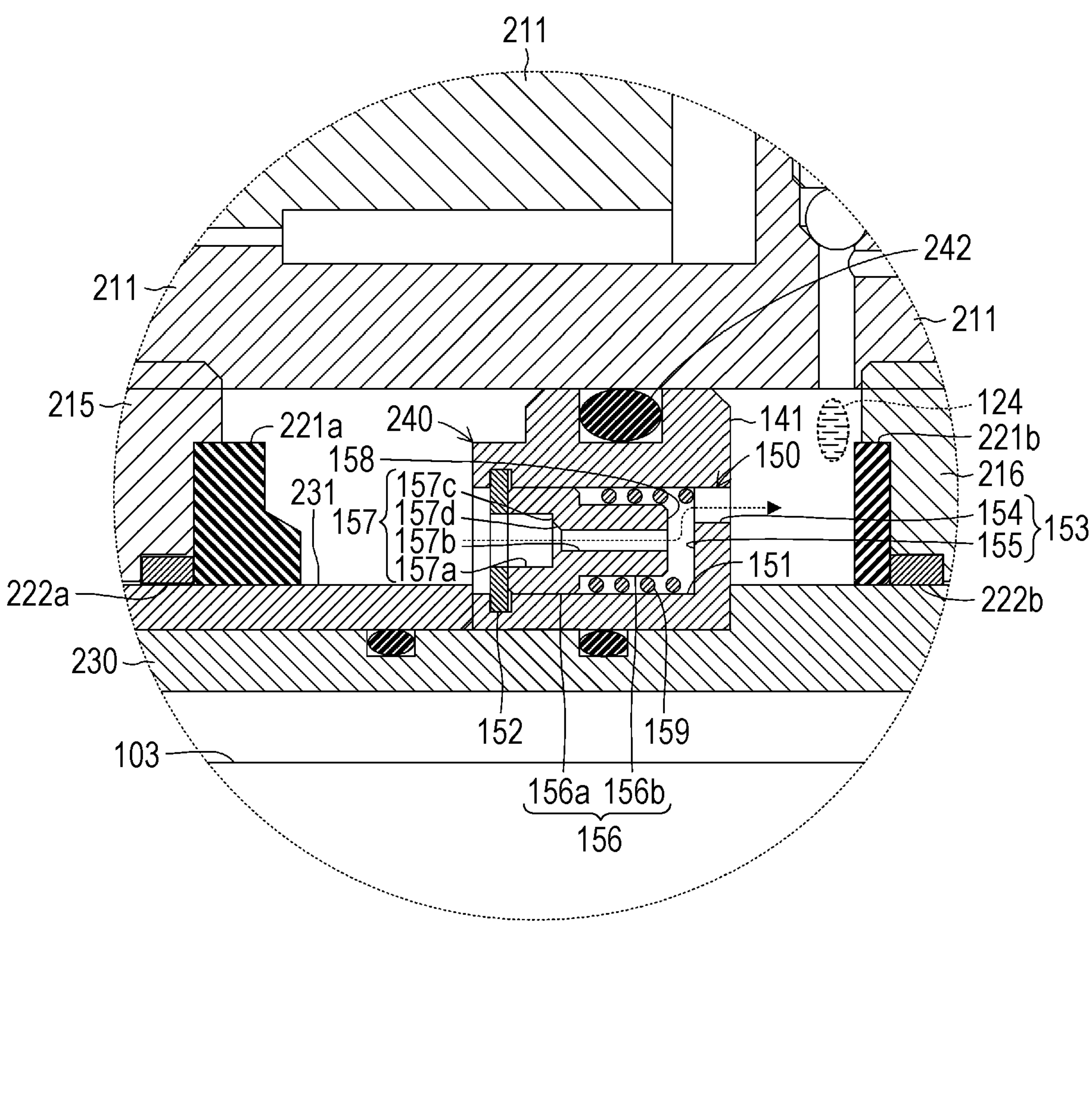
212

FIG. 12



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221b
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157c
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222a
222b
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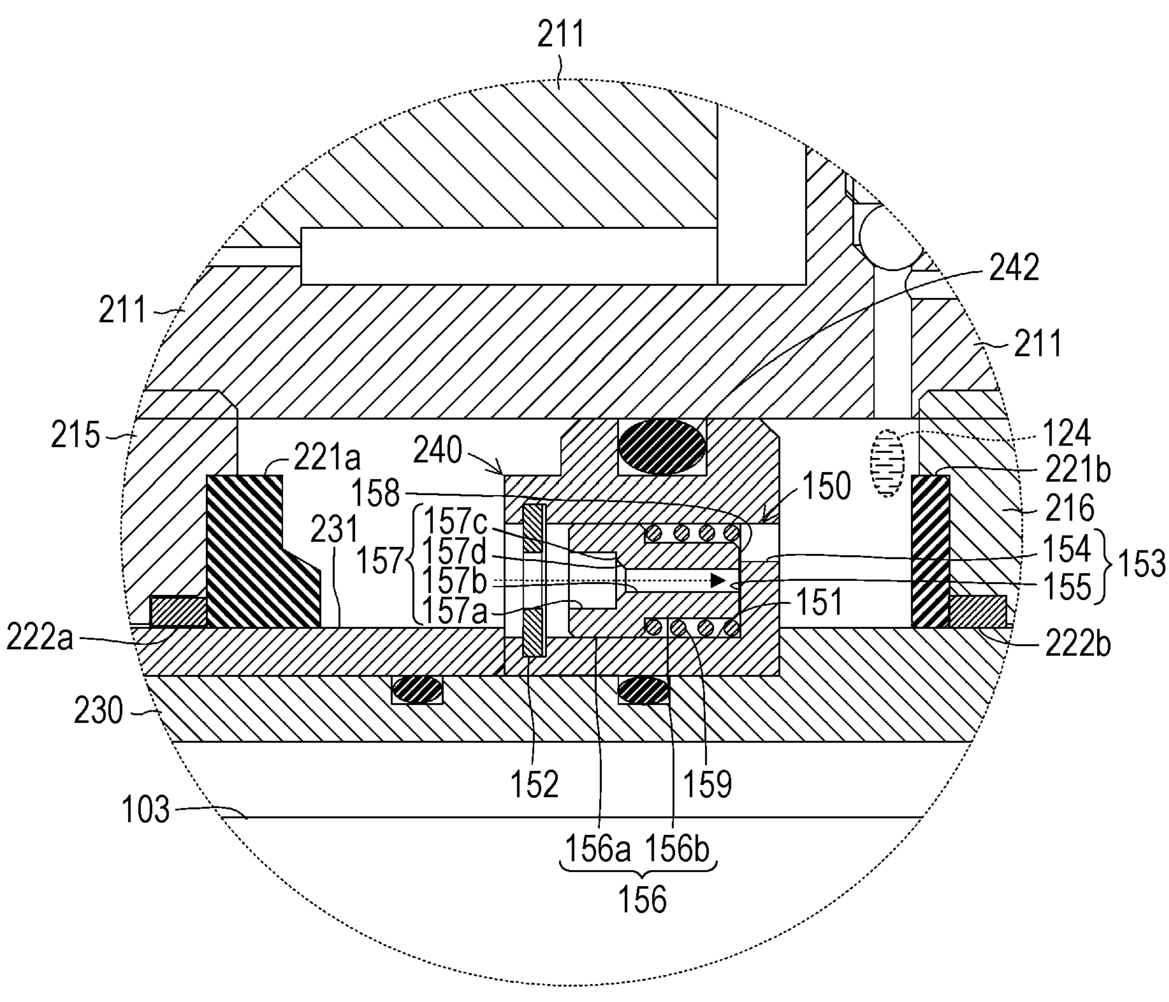
159
103
156a
156b
156

FIG. 13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156a
156b
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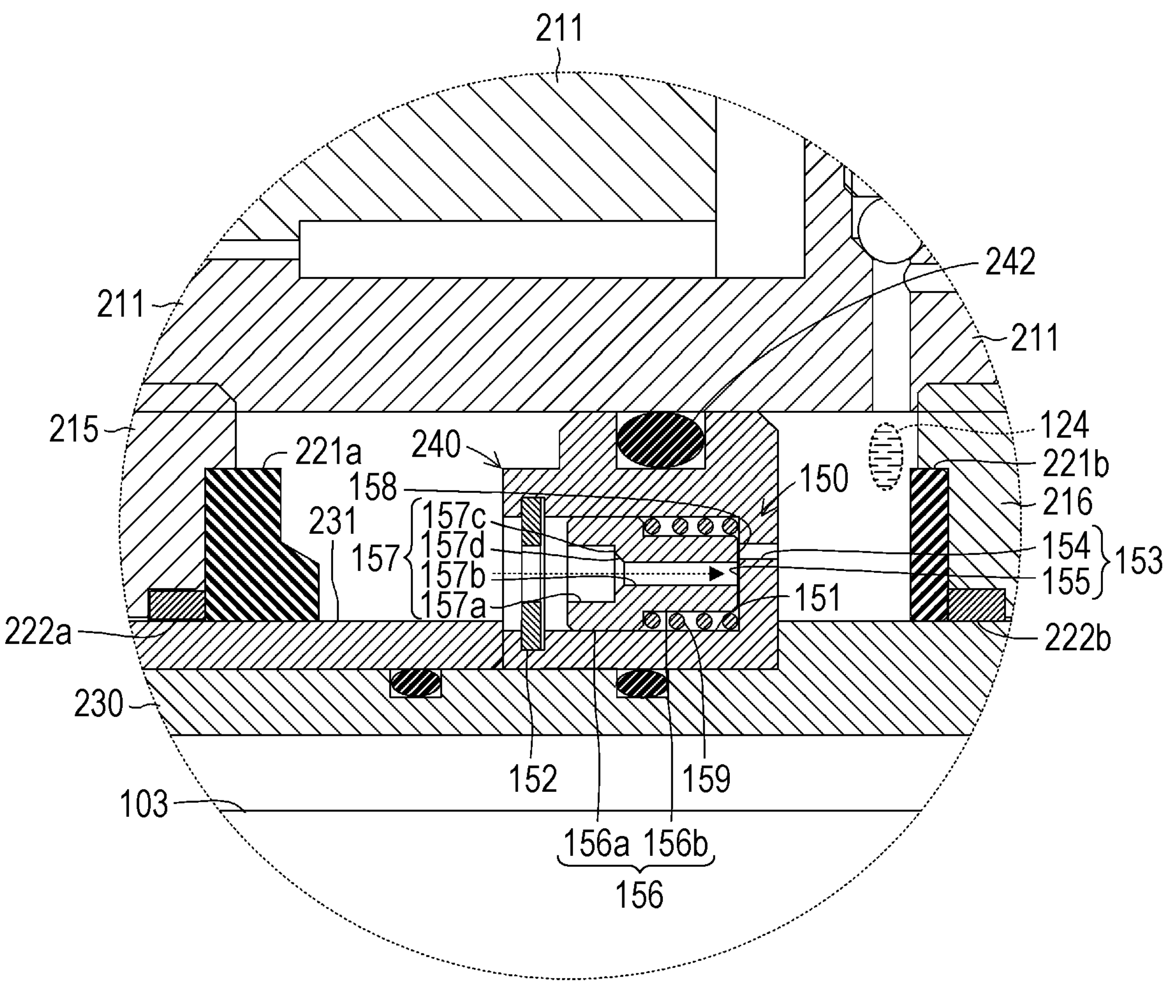
FIG. 14



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156a
156b
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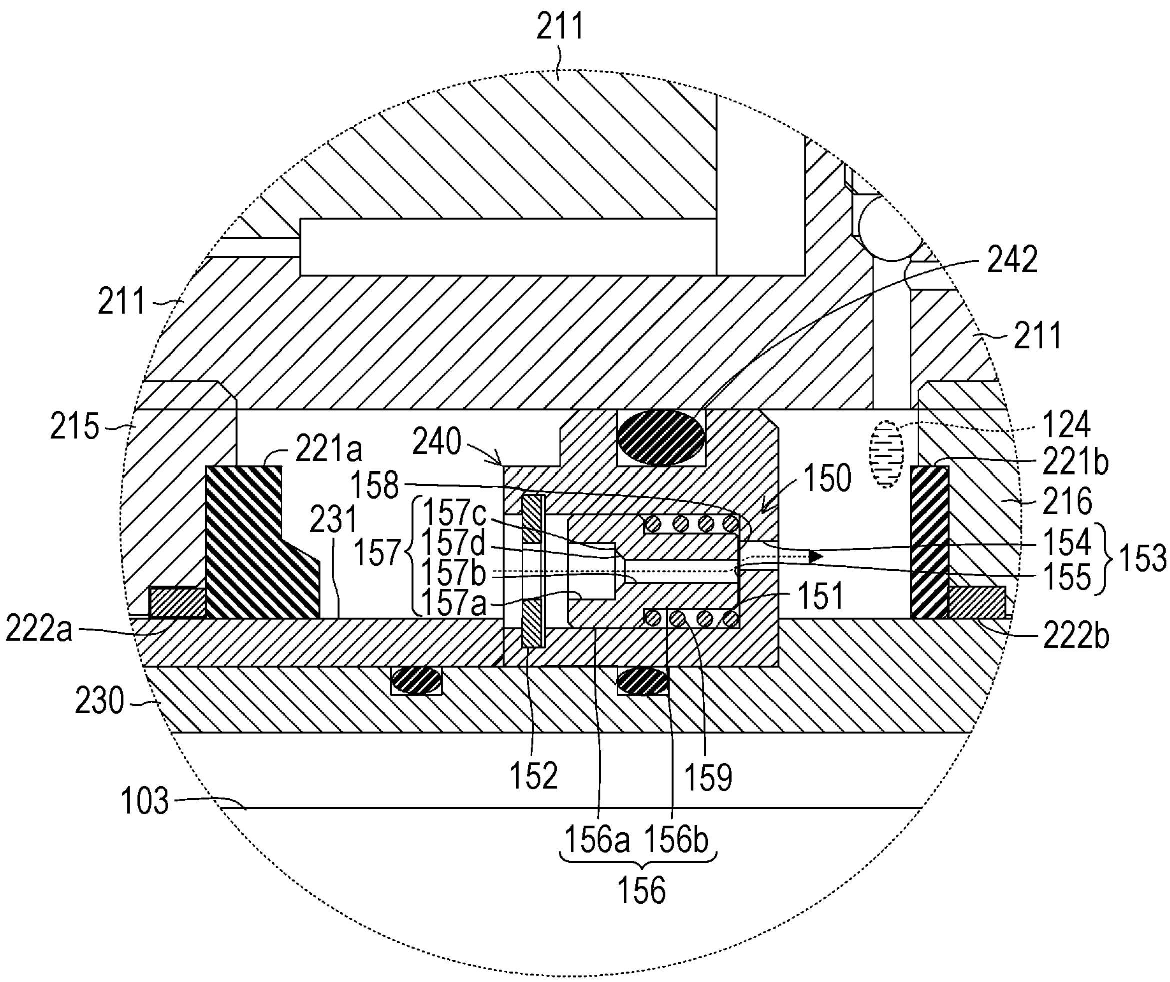
FIG. 15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156a
156b
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FIG. 16

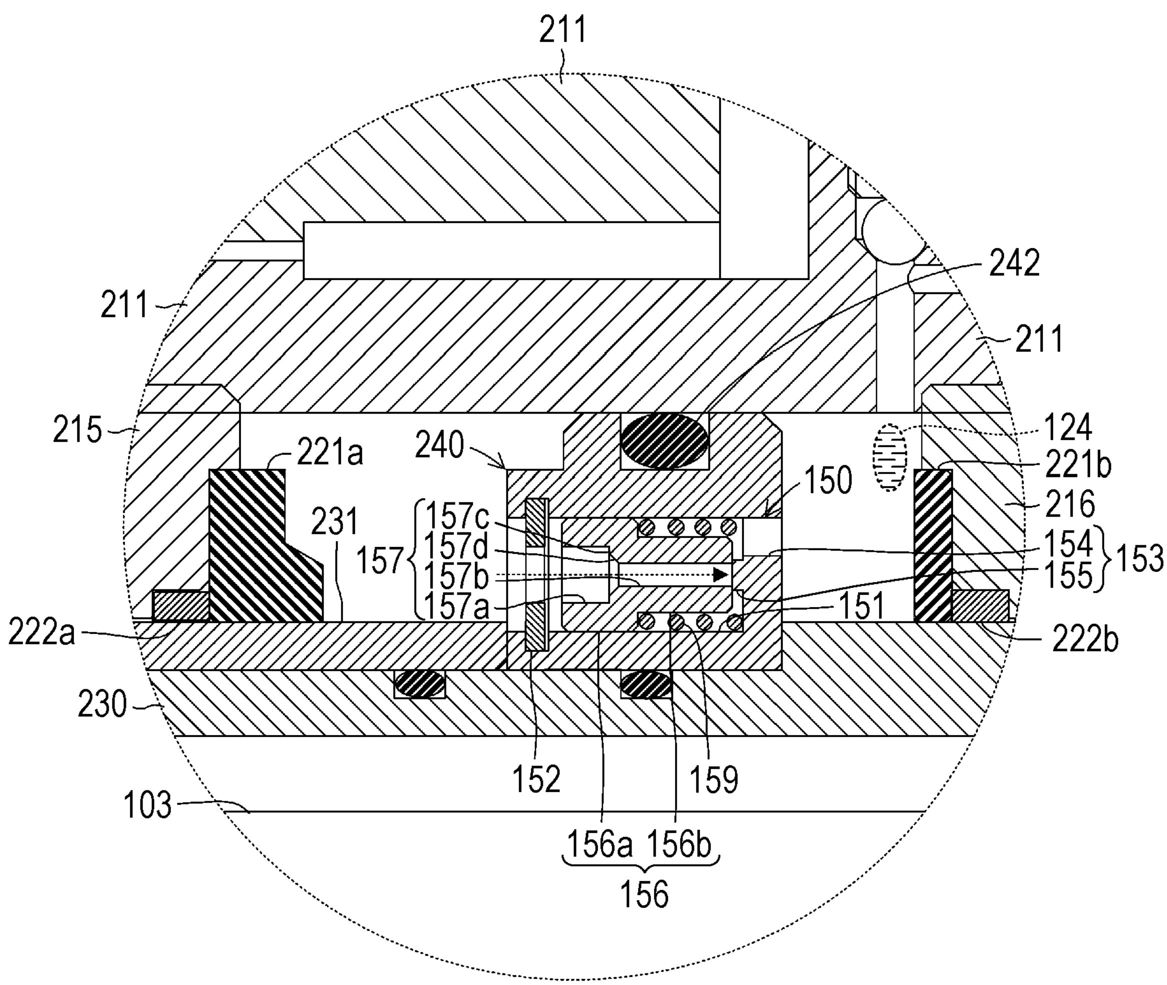
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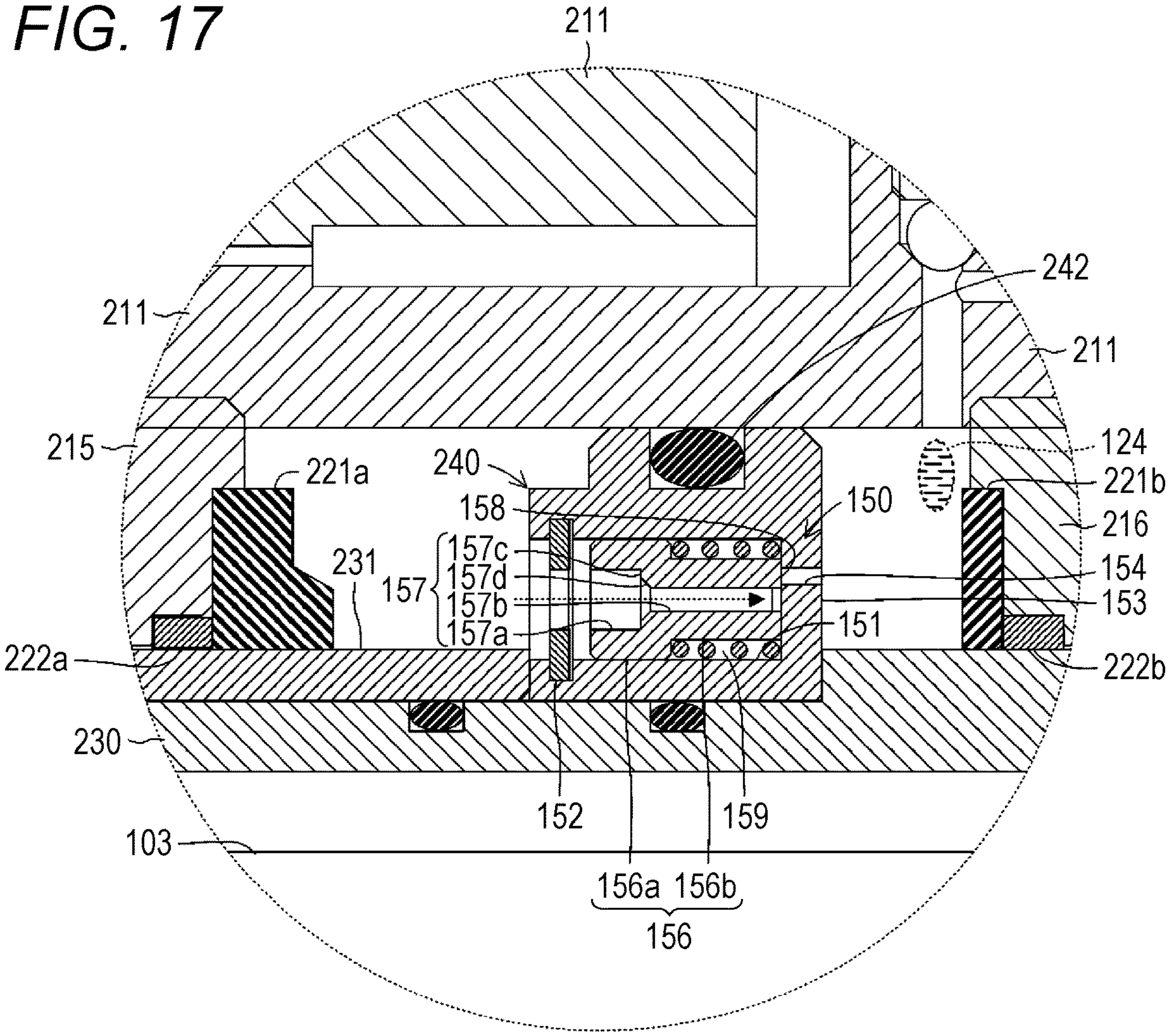
FIG. 17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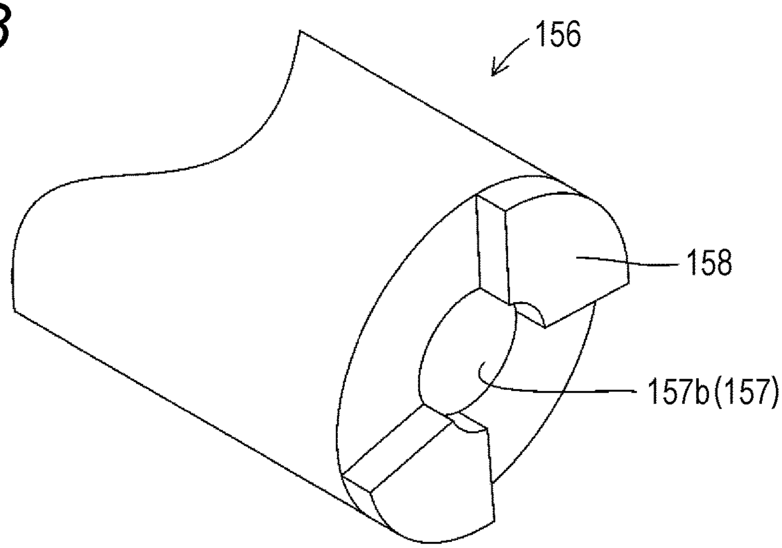
156a
156b
156

FIG. 18


156
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157b(157)

FLOW CONTROL VALVE, DAMPER AND STEERING DEVICE

TECHNICAL FIELD

The present invention relates to a flow control valve provided at a flow path, in which fluid flows, to control the flow of the fluid by causing the fluid to flow with a limitation on the flow of the fluid, a damper including the flow control valve, and a steering device including the damper.

BACKGROUND ART

Typically, in a four-wheeled self-propelled vehicle, a steering device is provided as a mechanical device for transmitting driver's steering operation to wheels to steer the self-propelled vehicle. In this case, in the steering device, an impact absorption member is provided to absorb strong impact generated, for example, when the self-propelled vehicle runs over a curbstone. For example, in a steering device disclosed in Patent Literature 1 below, an impact absorption member made of a rubber material and a metal material is provided between a rack housing covering a rack shaft and a tie rod connected to a wheel. With this configuration, strong impact generated, for example, when a self-propelled vehicle runs over a curbstone can be absorbed.

CITATION LIST

Patent Literature

PATENT LITERATURE 1: JP-A-2016-97840

SUMMARY OF INVENTION

However, in the steering device disclosed in Patent Literature 1 above, the impact absorption member is made of rubber or synthetic resin. For this reason, there is a problem that the magnitude of impact absorbable is small.

The present invention has been made in order to cope with the above-described problem. An object of the present invention is to provide a steering device capable of absorbing great impact force, a damper applicable to the steering device, and a flow control valve applicable to the damper.

In order to achieve the object described above, a feature of the present invention is a flow control valve provided at a flow path, in which fluid flows, to control a flow of the fluid by causing the fluid to flow with a limitation on the flow of the fluid. The flow control valve includes: a first flow body including a first flow hole through which the fluid flows; a second flow body arranged so as to face the first flow body and including a second flow hole through which the fluid flows; and a separation elastic body that produces an elastic force of separating the first flow body and the second flow body from a position at which the first flow body and the second flow body contact each other. In the flow control valve, at least one of the first flow body or the second flow body includes a hole diameter restriction portion that closes at least part of at least one of the second flow hole or the first flow hole when the first flow body and the second flow body contact each other.

According to the feature of the present invention configured as described above, in the flow control valve, at least one of the first flow body or the second flow body each having the first flow hole and the second flow hole in which the fluid flows includes the hole diameter restriction portion that closes at least part of at least one of the second flow hole or the first flow hole when the first flow body and the second flow body contact each other. With this configuration, in the flow control valve according to the present invention, in a case where great external force acts on between the first flow body and the second flow body, at least part of the first flow hole and the second flow hole is closed by the hole diameter restriction portion, and the flow of the fluid is limited accordingly. Thus, the steering device can be configured to damp the great external force and absorb great impact force.

In this case, in the flow control valve of the present invention, the hole diameter of the second flow hole and elastic force of the separation elastic body are selected as necessary so that a condition for closing the first flow hole and the second flow hole, i.e., the magnitude of damping force and a condition for generating the damping force, can be freely adjusted. Note that each hole diameter of the first flow hole and the second flow hole includes shapes other than a circular sectional shape (including an oval shape), such as a quadrangular shape, a polygonal shape, or various other irregular shapes.

Further, another feature of the present invention is the flow control valve described above, in which the hole diameter restriction portion is provided only at one of the first flow body or the second flow body.

According to another feature of the present invention configured as described above, in the flow control valve, the hole diameter restriction portion is provided only at one of the first flow body or the second flow body. Thus, the structure of the flow control valve and a step of manufacturing the flow control valve can be simplified.

Further, still another feature of the present invention is the flow control valve described above, in which the hole diameter restriction portion is provided at each of the first flow body and the second flow body.

According to still another feature of the present invention configured as described above, in the flow control valve, the hole diameter restriction portion is provided at each of the first flow body and the second flow body. Thus, reliability of control of the flow of the fluid can be improved.

Further, still another feature of the present invention is the flow control valve described above, in which the hole diameter restriction portion is formed so as to fully close at least one of the second flow hole or the first flow hole.

According to still another feature of the present invention configured as described above, in the flow control valve, the hole diameter restriction portion is formed so as to fully close at least one of the second flow hole or the first flow hole. Thus, the steering device can be configured to damp great external force and absorb great impact force.

Further, still another feature of the present invention is the flow control valve which includes a second flow body housing portion that movably houses the second flow body on a second flow body side with respect to the first flow body, in which the separation elastic body is provided between the first flow body and the second flow body in the second flow body housing portion.

According to still another feature of the present invention configured as described above, in the flow control valve, the second flow body is slidably housed in the second flow body housing portion. Thus, the second flow body can stably approach or separates from the first flow body.

Further, still another feature of the present invention is the flow control valve described above, in which the second flow body is, at an opening of the second flow hole on a side opposite to the first flow body, formed in such a tapered shape that a hole size decreases from an opening side toward a far side.

According to still another feature of the present invention configured as described above, in the flow control valve, the second flow body is, at the opening of the second flow hole on the side opposite to the first flow body, formed in such a tapered shape that the hole size decreases from the opening side toward the far side. Thus, as compared to a case where the second flow hole is formed straight, the fluid can easily flow into the second flow hole, and activation of the flow control valve can be stabilized. Moreover, in the flow control valve, the fluid can easily flow into the second flow hole, and a flow speed can be increased. Thus, the tapered portion receives strong pressing force from the fluid so that the second flow body can easily displace to a first flow body side.

Further, still another feature of the present invention is the flow control valve described above which further includes a one-way valve that causes the fluid to flow in a flow path different from the first flow body and the second flow body, in which the one-way valve allows the flow of the fluid from a first flow body side to a second flow body side, and blocks the flow of the fluid from the second flow body side to the first flow body side.

According to still another feature of the present invention configured as described above, the flow control valve includes, in addition to the first flow body and the second flow body, the one-way valve allowing the flow of the fluid from the first flow body side to the second flow body side and blocking the flow of the fluid from the second flow body side to the first flow body side. Thus, in the flow control valve according to the present invention, in a case where the fluid flows from the second flow body side to the first flow body side, the fluid actively flows in the second flow body so that the second flow body can easily displace. Moreover, in the flow control valve according to the present invention, in a case where the fluid flows from the first flow body side to the second flow body side, the fluid can more efficiently flow from the first flow body side to the second flow body side by the one-way valve in addition to the first flow body and the second flow body.

Further, still another feature of the present invention is the flow control valve described above which further includes a flow restriction valve that causes the fluid to flow with a limitation on the flow of the fluid in a flow path different from the first flow body and the second flow body, and the flow restriction valve causes the fluid to flow with the limitation on the flow of the fluid between the first flow body side and the second flow body side.

According to still another feature of the present invention configured as described above, the flow control valve includes, in addition to the first blow body and the second flow body, the flow restriction valve that causes the fluid to flow with the limitation on the flow of the fluid between the first flow body side and the second flow body side. Thus, when the fluid flows between the first flow body side and the second flow body side, the flow control valve having basic damping force can be configured in addition to the first flow body and the second flow body.

The present invention can be implemented not only as the invention relating to the flow control valve, but also as the invention relating to a damper including the flow control valve and a steering device including the damper.

Specifically, in a damper including an inner chamber forming body forming an inner chamber housing fluid in a liquid-tight manner and damping external force received by the fluid by limiting a flow of the fluid, it is preferred that the flow control valve according to any one of claims **1** to **8** is provided, the flow control valve causing the fluid to flow with the limitation on the flow of the fluid. According to this configuration, features and advantageous effects similar to those of the above-described flow control valve can be expected from the damper according to the present invention.

In this case, it is preferred that: the damper further includes a return elastic body that provides an elastic force of causing the fluid to flow from the first flow body side to the second flow body side in the flow control valve; the flow control valve is, in the inner chamber, provided displaceable relative to the inner chamber; and the return elastic body provides the elastic force to one of the inner chamber forming body or the flow control valve to displace the one of the inner chamber forming body or the flow control valve relative to the other one of the inner chamber forming body or the flow control valve.

According to this configuration, in the damper according to the present invention, the return elastic body provides the elastic force to one of the inner chamber forming body or the flow control valve to displace the one of the inner chamber forming body or the flow control valve relative to the other one of the inner chamber forming body or the flow control valve. With this configuration, in the damper according to the present invention, in a case where the fluid does not flow from the second flow body side to the first flow body side, the flow control valve is displaced relative to the inner chamber forming body so that the flow control valve can be constantly at a position at which the fluid can flow from the first flow body side to the second flow body side, i.e., an activation start position for producing a damping function of the flow control valve.

Further, in this case, it is preferred that: the damper further includes an integral displacement body that displaces integrally with the flow control valve relative to the inner chamber; the inner chamber forming body is formed in a solid bar shape or a tubular shape; the inner chamber is formed in a circular-ring tubular shape outside the inner chamber forming body; the flow control valve is formed at a ring-shaped valve support to be fitted in the circular-ring tubular inner chamber; and the integral displacement body is formed in a tubular shape slidably fitted onto the inner chamber forming body.

According to this configuration, in the damper according to the present invention, the inner chamber is formed outside the inner chamber forming body. Thus, the inner chamber forming body itself can be formed in the solid bar shape or the tubular shape, and the number of variations in damper installation can be increased. Note that the phrase of the inner chamber being formed in the circular-ring tubular shape means a circular-ring tubular shape formed with a circular ring-shaped sectional shape and extending in a tubular body shape.

Further, in this case, it is preferred that: the damper further includes an integral displacement body that displaces integrally with the flow control valve relative to the inner chamber; the inner chamber forming body is formed in a tubular shape; the inner chamber is formed in a circular-ring tubular shape inside the inner chamber forming body; the flow control valve is formed at a ring-shaped valve support to be fitted in the circular-ring tubular inner chamber; and the integral displacement body is formed in a solid bar shape or a tubular shape to be slidably fitted in the inner chamber forming body.

According to this configuration, in the damper according to the present invention, the inner chamber is formed inside the inner chamber forming body. Thus, the integral displacement body itself can be formed in the solid bar shape or the tubular shape, and the number of variations in damper installation can be increased.

Further, specifically, it is only required that the steering device, which includes a steering shaft formed so as to extend in a bar shape and rotated by operation of a steering wheel, a rack bar formed so as to extend in a rod shape and converting rotary motion of the steering shaft into reciprocating motion in an axis direction to transmit the reciprocating motion, an intermediate coupling body coupled to each end portion of the rack bar to directly or indirectly couple a wheel targeted for steering to the each end portion, and a rack housing covering the rack bar, is configured such that the steering device includes the damper according to any one of claims **9** to **12**, and the damper is provided between the rack housing and the rack bar or the intermediate coupling body to absorb impact from the wheel. According to this configuration, features and advantageous effects similar to those of the above-described flow control valve and the above-described damper can be expected from the steering device according to the present invention.

In this case, it is preferred that: the steering device includes the damper according to claim **11**; the inner chamber forming body is formed at the intermediate coupling body; and the integral displacement body is formed at such a position that the integral displacement body contacts or separates from the rack housing by the reciprocating motion of the rack bar.

According to this configuration, in the steering device according to the present invention, the inner chamber forming body is formed at the intermediate coupling body such as a tie rod or a rack end, and the integral displacement body is formed at such a position that the integral displacement body contacts or separates from the rack housing by the reciprocating motion of the rack bar in the inner chamber forming body. That is, in the steering device according to the present invention, the damper is provided at the intermediate coupling body such as the tie rod or the rack end so that a damper maintenance or replacement process can be easily performed.

In this case, it is preferred that: the steering device includes the damper according to claim **12**; the inner chamber forming body is formed at an end portion of the rack housing; and the rack bar or the intermediate coupling body penetrates the integral displacement body; and the integral displacement body is formed at such a position that the rack bar or the intermediate coupling body contacts or separates from the integral displacement body by the reciprocating motion of the rack bar.

According to this configuration, in the steering device according to the present invention, the inner chamber forming body is formed at the end portion of the rack housing, the rack bar or the intermediate coupling body (e.g., the tie rod or the rack end) penetrates the integral displacement body, and the integral displacement body is formed at such a position that the rack bar or the tie rod contacts or separates from the integral displacement body by the reciprocating motion of the rack bar. Thus, in the steering device according to the present invention, the damper is provided at the rack housing so that the intermediate coupling body such as the tie rod or the rack end can be decreased in weight.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a schematic view for describing the outline of an entire configuration of a steering device according to a first embodiment of the present invention.

FIG. **2** is a perspective view showing the outline of an external configuration of a damper according to the first embodiment of the present invention, the damper forming the steering device shown in FIG. **1**.

FIG. **3** is a sectional view showing the outline of an internal configuration of the damper shown in FIG. **2**.

FIG. **4** is a sectional view of an integral displacement body and a socket main body along a **4-4** line shown in FIG. **3**.

FIG. **5** is a partially-enlarged view showing details of the structure of the damper of FIG. **3** in a dashed circle **5**.

FIG. **6** is a sectional view showing the state of the damper shown in FIG. **3** at a moment when the integral displacement body has contacted a rack housing.

FIG. **7** is a partially-enlarged view showing the state of the damper shown in FIG. **5** when a first flow control valve causes fluid to flow.

FIG. **8** is a partially-enlarged view showing the state of the damper shown in FIG. **5** when the first flow control valve does not cause the fluid to flow.

FIG. **9** is a sectional view showing the outline of an internal configuration of a damper according to a second embodiment of the present invention.

FIG. **10** is a front view of only a flow control valve along a **10-10** line shown in FIG. **9**.

FIG. **11** is a sectional view showing the state of the damper shown in FIG. **9** at a moment when a socket main body has contacted an integral displacement body.

FIG. **12** is a partially-enlarged view of a portion, which is indicated by a dashed circle **12**, of the damper shown in FIG. **9** in a state in which a first flow control valve causes fluid to flow.

FIG. **13** is a partially-enlarged view showing the state of the damper shown in FIG. **9** when the first flow control valve does not cause the fluid to flow.

FIG. **14** is a partially-enlarged view showing the state of a damper according to a variation of the present invention when both a first flow hole and a second flow hole are fully closed such that a first flow control valve does not cause fluid to flow.

FIG. **15** is a partially-enlarged view showing the state of a damper according to another variation of the present invention when part of a first flow hole and part of a second flow hole overlap with each other such that the flow of fluid is ensured.

FIG. **16** is a partially-enlarged view showing the state of a damper of still another variation of the present invention when only a second flow hole is fully closed such that a first flow control valve does not cause fluid to flow.

FIG. **17** is a partially-enlarged view showing the state of a damper according to still another variation of the present invention when only a first flow hole is fully closed such that a first flow control valve does not cause fluid to flow.

FIG. **18** is a partially-enlarged perspective view showing an external configuration of a tip end portion of a second flow body shown in FIG. **17**.

DESCRIPTION OF EMBODIMENTS

First Embodiment

Hereinafter, a first embodiment as one embodiment of a steering device including flow control valves and dampers according to the present invention will be described with reference to the drawings. FIG. **1** is a schematic view for describing the outline of an entire configuration of a steering device **100** according to the first embodiment of the present invention. FIG. **2** is a perspective view showing the outline of an external configuration of a damper **120** according to the first embodiment of the present invention, the damper **120** forming the steering device **100** shown in FIG. **1**. FIG. **3** is a sectional view showing the outline of an internal configuration of the damper **120** shown in FIG. **2**. FIG. **4** is a sectional view of an integral displacement body **130** and a socket main body **107** along a **4-4** line shown in FIG. **3**. FIG. **5** is a partially-enlarged view showing details of the structure of the damper **120** of FIG. **3** in a dashed circle **5**.

The steering device **100** is a mechanical device for steering two front wheels (or rear wheels) of a four-wheeled self-propelled vehicle (not shown) right and left.

(Configuration of Steering Device **100**)

The steering device **100** includes a steering wheel **101**. The steering wheel **101** is an operator (i.e., a handle) for manually operating a travelling direction by a driver of the self-propelled vehicle. The steering wheel **101** is formed in such a manner that a resin material or a metal material is formed in a circular ring shape. A steering shaft **102** is coupled to the steering wheel **101**.

The steering shaft **102** is a component formed in a bar shape and rotating about an axis according to clockwise or counterclockwise rotary operation of the steering wheel **101**. The steering shaft **102** is formed in such a manner that one or more metal bar bodies are coupled through, e.g., a universal joint. The steering wheel **101** is coupled to one end portion of the steering shaft **102**, and a pinion gear **102***a* is formed at the other end portion and a rack bar **103** is coupled to the other end portion.

The rack bar **103** is a component formed in a bar shape and reciprocatably displacing in an axis direction to transmit a force of steering each of two wheels **112** and an amount of steering each of two wheels **112** to a corresponding one of knuckle arms **111**. The rack bar **103** is made of a metal material. In this case, a rack gear **103***a* is formed at part of the rack bar **103**, and engages with the pinion gear **102***a* of the steering shaft **102**. That is, the pinion gear **102***a* and the rack gear **103***a* form a rack-and-pinion mechanism (a steering gear box) converting rotary motion of the steering shaft **102** into linear reciprocating motion of the rack bar **103**.

Both end portions of the rack bar **103** in the axis direction are exposed through a rack housing **104** in a state in which the rack-and-pinion mechanism is covered with the rack housing **104**. The wheels **112** are each coupled to both end portions of the rack bar **103** exposed through the rack housing **104** through intermediate coupling bodies **105** and the knuckle arms **111**.

The rack housing **104** is a component for covering and protecting a main portion of the rack bar **103**, such as the rack-and-pinion mechanism. The rack housing **104** is formed in such a manner that a metal material is formed in a cylindrical shape. The rack housing **104** is attached in a fixed manner to a chassis (not shown) of the self-propelled vehicle.

The intermediate coupling body **105** is a component for transmitting the steering force and the steering amount transmitted from the rack bar **103** to the knuckle arm **111**. The intermediate coupling body **105** mainly includes a rack end **106** and a tie rod **110**. The rack end **106** is a component movably coupling the tie rod **110** to a tip end portion of the rack bar **103** and formed with the damper **120**. The rack end **106** mainly includes the socket main body **107** and a stud body **108**

The socket main body **107** is a component movably coupling the stud body **108** to the tip end portion of the rack bar **103** and formed with the damper **120**. The socket main body **107** is formed in such a manner that a metal material is formed in a round bar shape. A ball holding portion **107***a* is formed at one (the right side as viewed in the figure) end portion of the socket main body **107**, and an external thread portion **107***b* to be fitted in the tip end portion of the rack bar **103** is formed at the other (the left side as viewed in the figure) end portion. The ball holding portion **107***a* is formed in such a recessed spherical shape that the ball holding portion **107***a* is slidably fitted onto a ball portion **108***a* of the stud body **108** to hold the ball portion **108***a*. The damper **120** is formed between the ball holding portion **107***a* and the external thread portion **107***b* of the socket main body **107**.

The stud body **108** is a component for movably coupling the tie rod **110** to the socket main body **107**. The stud body **108** is formed in such a manner that a metal material is formed in a round bar shape. The spherical ball portion **108***a* is formed at one (the left side as viewed in the figure) end portion of the stud body **108**, and an external thread portion (not shown) to be fitted in an end portion of the tie rod **110** is formed at the other (the right side as viewed in the figure) end portion.

The tie rod **110** is a component movably coupling the knuckle arm **111** to a tip end portion of the rack end **106**. The tie rod **110** is formed in such a manner that a ball joint is movably attached to a tip end portion of a tie rod main body extending in a bar shape. The knuckle arm **111** is a metal component for holding the wheel **112** on the tie rod **110** to transmit the steering force and the steering amount transmitted from the tie rod **110** to the wheel **112**. The knuckle arm **111** is formed in such a shape that multiple bar-shaped bodies extend from the periphery of a cylindrical portion. The wheels **112** are a pair of right and left components rolling on a rod surface such that the self-propelled vehicle moves back and forth. The wheel **112** is formed in such a manner that a rubber tire is attached to the outside of a metal wheel.

The damper **120** is a tool for absorbing strong pressing force (impact) transmitted from the wheel **112**. The damper **120** is formed at each of the right and left intermediate coupling bodies, more specifically each of the right and left socket main bodies **107**. The damper **120** includes an inner chamber **121**.

The inner chamber **121** is a portion where fluid **124** is housed in a liquid-tight manner. The inner chamber **121** is, on an outer peripheral portion of the socket main body **107**, formed in a circular-ring tubular shape cut out in a recessed shape along a circumferential direction and extending in the axis direction. That is, the socket main body **107** is equivalent to an inner chamber forming body according to the present invention. Of the inner chamber **121**, each of a bottom portion and one (the right side as viewed in the figure) end portion in the axis direction of the socket main body **107** is, in the present embodiment, formed by the socket main body **107** itself, and the other (the left side as viewed in the figure) end portion in the axis direction is formed by a wall forming body **122**. The outside of the inner chamber **121** is covered with the integral displacement body **130**.

The wall forming body **122** is a component for forming the left wall portion of the inner chamber **121** as viewed in the figure. The wall forming body **122** is formed in such a manner that a metal material is formed in a circular ring shape. The wall forming body **122** is fitted onto an outer peripheral surface of the socket main body **107** on the other (the left side as viewed in the figure) side in the axis direction, and is integrated with the socket main body **107**. At both end portions of the inner chamber **121** in the axis direction of the socket main body **107**, buffers **123***a*, **123***b* made of elastomer such as urethane resin and including elastic bodies are provided. In this case, the buffer **123***a* is formed thicker than the buffer **123***b*.

The fluid **124** is a substance for providing resistance to a flow control valve **140** sliding in the inner chamber **121** such that the damper function of the damper **120** works. The inner chamber **121** is filled with the fluid **124**. The fluid **124** includes a liquid, gel, or semisolid substance having viscosity according the specifications of the damper **120** and fluidity. In this case, the viscosity of the fluid **124** is selected as necessary according to the specifications of the damper **120**. In the present embodiment, the fluid **124** includes oil such as mineral oil or silicone oil. Note that the fluid **124** is hatched in a dashed circle in FIGS. **3** and **5** (the same also applies to FIGS. **9** and **12** to **15**).

Sliding bushes **125***a*, **125***b* are each fitted onto the outer peripheral surfaces of the socket main body **107** and the wall forming body **122** on both sides of the inner chamber **121** in the axis direction of the socket main body **107**. The sliding bush **125***a*, **125***b* is a component for smoothly reciprocatably sliding the integral displacement body **130** in the axis direction of the socket main body **107**. The sliding bush **125***a*, **125***b* is formed in such a manner that a metal material is formed in a circular ring shape having a slightly-larger outer diameter than the outer diameter of the socket main body **107**.

Seal rings **126***a*, **126***b* made of an elastomer material such as a rubber material and including elastic bodies are each fitted onto the outer peripheral surfaces of the socket main body **107** and the wall forming body **122** on the opposite side of these sliding bushes **125***a*, **125***b* from the inner chamber **121**. These seal rings **126***a*, **126***b* prevent leakage of the fluid **124** from the inner chamber **121** when the integral displacement body **130** slidably displaces relative to the socket main body **107**.

The integral displacement body **130** is a component covering the outside of the inner chamber **121** in a radial direction and formed with the flow control valve **140**. The integral displacement body **130** is formed in such a manner that a metal material is formed in a cylindrical shape. Specifically, the integral displacement body **130** is formed in such a cylindrical shape that the integral displacement body **130** is slidably fitted onto the outer peripheral surface of the socket main body **107**. In this case, the integral displacement body **130** is formed with such a length that the integral displacement body **130** projects from a rack-housing-**104**-side end portion of the socket main body **107**.

The flow control valve **140** is formed so as to project from a center portion of an inner peripheral portion of the integral displacement body **130** in the axis direction. An elastic body holding portion **131** is formed between the flow control valve **140** and an inner peripheral surface of the integral displacement body **130**. A dust boot **133** and a dust seal **134** are each provided at both end portions of the inner peripheral portion of the integral displacement body **130** in the axis direction.

The elastic body holding portion **131** is a portion housing one of both end portions of a return elastic body **132**. The elastic body holding portion **131** is formed in a circular ring shape between the inner peripheral surface of the integral displacement body **130** and the flow control valve **140**. The return elastic body **132** is a component for elastically pressing the flow control valve **140** to the left end portion of the inner chamber **121** as viewed in the figure. The return elastic body **132** includes a metal coil spring. One (the left side as viewed in the figure) end portion of the return elastic body **132** is housed in the elastic body holding portion **131** to elastically press the integral displacement body **130**, and the other (the right side as viewed in the figure) end portion elastically presses the outer peripheral portion of the socket main body **107** through the sliding bush **125***a*.

The dust boot **133** is a component for preventing dust from entering the integral displacement body **130** from a rack housing **104** side which is one (the left side as viewed in the figure) of both end portions of the integral displacement body **130**. The dust boot **133** is formed in such a manner that an elastomer material such as a rubber material is formed in a cylindrical shape. One end portion of the dust boot **133** is connected to the end portion of the integral displacement body **130**, and the other end portion is connected to the wall forming body **122**. Note that the dust boot **133** is not shown in FIG. **2**.

The dust seal **134** is, as in the dust boot **133**, a component preventing dust from entering the integral displacement body **130** from a stud body **108** side which is the other (the right side as viewed in the figure) one of both end portions of the integral displacement body **130**. The dust seal **134** is formed in such a manner that an elastomer material such as a rubber material is formed in a circular ring shape. The dust seal **134** is fitted in a groove cut out in a circular ring shape at the end portion of the integral displacement body **130**.

An accumulator housing portion **135** is formed at an outer peripheral portion of the integral displacement body **130**. The accumulator housing portion **135** is a tubular portion for housing an accumulator **136** in a liquid-tight manner. The accumulator housing portion **135** is formed so as to project onto an outer peripheral surface of the integral displacement body **130** and extend in a longitudinal direction of the integral displacement body **130**. One end portion of the accumulator housing portion **135** communicates with the inner chamber **121** on a first flow body **153** side, and the other end portion is sealed with a plug.

The accumulator **136** is a tool compensating for a volume change in the fluid **124** in the inner chamber **121** due to expansion or contraction caused by a temperature change. The accumulator **136** is configured to house a piston reciprocatably sliding in the accumulator housing portion **135** in a state in which the piston is elastically pressed to an inner chamber **121** side by a coil spring.

The flow control valve **140** is a tool causing the fluid **124** to flow with a limitation on the flow of the fluid **124** in the inner chamber **121** to control the flow of the fluid **124**, thereby generating damping force of the damper **120**. The flow control valve **140** mainly includes each of a valve support **141**, a first flow control valve **150**, a second flow control valve **160**, and third flow control valves **170**.

The valve support **141** is a portion formed with each of the first flow control valve **150**, the second flow control valve **160**, and the third flow control valves **170**. The valve support **141** is formed in a flat-plate circular ring shape projecting inward from the inner peripheral portion of the integral displacement body **130**. An inner peripheral portion of the valve support **141** is formed with a smooth cylindrical surface such that the valve support **141** slides on the bottom portion of the inner chamber **121** in a liquid-tight manner, and a seal ring **142** including an elastic body is fitted in such an inner peripheral portion.

The first flow control valve **150** is a valve functioning as a trigger for generating the maximum damping force when strong impact force acts on the damper **120**. The first flow control valve **150** mainly includes each of a second flow body housing portion **151**, a first flow body **153**, a second flow body **156**, and a separation elastic body **159**. The second flow body housing portion **151** is a portion slidably housing the later-described second flow body **156**. The second flow body housing portion **151** is formed in a bottomed tubular shape constantly opened at one (the right side as viewed in the figure) end portion of the valve support **141**.

In this case, the second flow body housing portion **151** is formed so as to be constantly opened at one of two side surfaces of the valve support **141** on the front side when the integral displacement body **130** slidably displaces against elastic force of the return elastic body **132** relative to the socket main body **107**. A retaining ring **152** is fitted in a ring-shaped groove formed close to an opening at an inner peripheral surface of the second flow body housing portion **151**. The retaining ring **152** is a component for preventing detachment of the second flow body **156** housed in the second flow body housing portion **151**. The retaining ring **152** includes a metal C-shaped ring body.

The first flow body **153** is a portion for controlling the flow of the fluid **124** in cooperation with the second flow body **156**. The first flow body **153** mainly includes each of a first flow hole **154** and a first hole diameter restriction portion **155**. The first flow hole **154** is a through-hole causing the fluid **124** to flow, and is formed at a bottom portion of the second flow body housing portion **151**. In this case, the first flow hole **154** is formed at an edge portion of the bottom portion eccentric with respect to the center line of the second flow body housing portion **151**. That is, the second flow body housing portion **151** is opened larger on one side (a buffer **123***a* side) of the inner chamber **121**, and due to the first flow hole **154**, is opened smaller on the other side (a buffer **123***b* side).

The first hole diameter restriction portion **155** is a portion blocking the flow of the fluid **124** in a second flow hole **157**. The first hole diameter restriction portion **155** is formed in a wall shape at the periphery of the first flow hole **154**. Moreover, the first hole diameter restriction portion **155** is formed at a position facing the second flow hole **157** so as to fully close the second flow hole **157** of the second flow body **156** when the second flow body **156** has contacted the first flow body **153**. In the present embodiment, the first hole diameter restriction portion **155** is formed by the bottom portion of the second flow body housing portion **151**.

The second flow body **156** is a component for controlling the flow of the fluid **124** in cooperation with the first flow body **153**. The second flow body **156** is formed in such a manner that a metal material is formed in a cylindrical shape. In this case, the second flow body **156** includes a large-diameter portion **156***a* sliding on the inner peripheral surface of the second flow body housing portion **151** and a small-diameter portion **156***b* having a smaller diameter than that of the large-diameter portion **156***a*. Each of the second flow hole **157** and a second hole diameter restriction portion **158** is formed at the second flow body **156**.

The second flow hole **157** is a through-hole causing the fluid **124** to flow. The second flow hole **157** includes a large-diameter hole **157***a* and a small-diameter hole **157***b* penetrating the second flow body **156**. The large-diameter hole **157***a* is, at the second flow body **156**, formed so as to be opened at the front surface when the integral displacement body **130** slidably displaces against the elastic force of the return elastic body **132**. The small-diameter hole **157***b* extends from the farthest portion of the large-diameter hole **157***a* to the rear surface, and is opened when the integral displacement body **130** slidably displaces against the elastic force of the return elastic body **132**.

In this case, a flat-plate annular step portion **157***c* is formed between the large-diameter hole **157***a* and the small-diameter hole **157***b*. The small-diameter hole **157***b* is formed with a tapered portion **157***d* in a tapered shape, and at the tapered portion **157***d*, the hole diameter continuously decreases from an end portion on a large-diameter hole **157***a* side toward the far side of the small-diameter hole **157***b*. The small-diameter hole **157***b* is formed at such a position with such a size that the small-diameter hole **157***b* communicates with the large-diameter hole **157***a* and faces the first hole diameter restriction portion **155** without facing the first flow hole **154**. That is, the small-diameter hole **157***b* is formed at such a position with such a size that, on the first hole diameter restriction portion **155**, the small-diameter hole **157***b* does not overlap with the first flow hole **154**. In the present embodiment, the small-diameter hole **157***b* is formed concentrically with the second flow body **156** and the large-diameter hole **157***a*, and is formed with a smaller diameter than that of the first flow hole **154**.

The second hole diameter restriction portion **158** is a portion blocking the flow of the fluid **124** in the first flow hole **154**. The second hole diameter restriction portion **158** is formed in a wall shape at the periphery of the small-diameter hole **157***b* forming the second flow hole **157**. Specifically, the second hole diameter restriction portion **158** is formed at a position facing the first flow hole **154** so as to partially close the first flow hole **154** of the first flow body **153** when the second flow body **156** has contacted the first flow body **153**. In the present embodiment, the second hole diameter restriction portion **158** is formed in a flat-plate annular shape with such a size that the second hole diameter restriction portion **158** closes about ⅓ of the first flow hole **154** of the first flow body **153** when the second flow body **156** has contacted the first flow body **153**.

The separation elastic body **159** is a component producing an elastic force of separating the second flow body **156** from the first flow body **153** in the second flow body housing portion **151**. The separation elastic body **159** includes a metal coil spring. One (the left side as viewed in the figure) end portion of the separation elastic body **159** presses the first hole diameter restriction portion **155** (the bottom portion of the second flow body housing portion **151**), and the other (the right side as viewed in the figure) end portion is fitted onto an outer peripheral portion of the small-diameter portion **156***b*. The elastic force of the separation elastic body **159** is set to a level corresponding to the magnitude of external force for which the damper **120** needs to generate the maximum damping force. One first flow control valve **150** is provided at the valve support **141**.

The second flow control valve **160** is a valve blocking the flow of the fluid **124** from the front side to the rear side in slide displacement when the integral displacement body **130** slidably displaces against the elastic force of the return elastic body **132** relative to the socket main body **107**, and is also a valve allowing the fluid **124** to easily flow from the front side to the rear side in slide displacement when the integral displacement body **130** slidably displaces by the elastic force of the return elastic body **132**. That is, the second flow control valve **160** includes a one-way valve. The configuration of the one-way valve as the second flow control valve **160** is well-known, and therefore, detailed description thereof will be omitted. One second flow control valve **160** is provided at a position of 180° from the first flow control valve **150** in the circumferential direction at the valve support **141**.

The third flow control valve **170** is a valve causing the fluid **124** to flow with the limitation on the flow of the fluid

124 when the integral displacement body **130** slidably displaces against the elastic force of the return elastic body **132** relative to the socket main body **107** and when the integral displacement body **130** slidably displaces by the elastic force of the return elastic body **132**. The third flow control valve **170** includes a thin through-hole formed at the valve support **141**. In the present embodiment, the third flow control valves **170** are each formed at two intermediate positions between the first flow control valve **150** and the second flow control valve **160** in the circumferential direction at the valve support **141**. Note that the third flow control valve **170** is equivalent to a flow restriction valve according to the present invention.

(Actuation of Steering Device **100**)

Next, actuation of the steering device **100** configured as described above will be described. The steering device **100** is incorporated as the mechanism that steers the steered wheels (e.g., two front wheels) of the not-shown four-wheeled self-propelled vehicle right and left into the self-propelled vehicle. The steering device **100** changes the direction of each of two wheels **112** according to operation of the steering wheel **101** by the driver of the self-propelled vehicle, thereby determining the travelling direction of the self-propelled vehicle.

During travelling of such a self-propelled vehicle, each damper **120** in the steering device **100** is activated when the rack bar **103** has displaced to the vicinity of a right or left displacement limit with respect to the pinion gear **102***a*. In this case, the displacement limit of the rack bar **103** is a right or left steering limit of the wheel **112**, and for example, includes not only a case where the driver of the self-propelled vehicle has turned the steering wheel **101** clockwise or counterclockwise to the vicinity of a turning limit, but also a case where great input acts on the rack bar **103** from a wheel **112** side due to collision of the wheel **112** with an obstacle such as a curbstone.

First, a case where the damper **120** is not activated will be described. As shown in FIG. **3**, the integral displacement body **130** does not contact the rack housing **104** within an area where the rack bar **103** does not reach the vicinity of the displacement limit, such as a case where the wheel **112** of the self-propelled vehicle is not steered to the vicinity of the steering limit, and therefore, the damper **120** is not activated. In this case, in the damper **120**, the flow control valve **140** is pressed against the wall forming body **122** in the inner chamber **121** through the buffer **123***b* by the elastic force of the return elastic body **132**, as shown in FIG. **5**. That is, the integral displacement body **130** is maintained in a state in which the integral displacement body **130** is elastically positioned at a position closest to the rack housing **104** side on the socket main body **107**.

The first flow control valve **150** is maintained in a state in which the second flow body **156** is positioned at a position farthest from the first flow body **153** by the elastic force of the separation elastic body **159**. That is, the first flow control valve **150** is in a state in which the flow of the fluid **124** is allowed.

Next, a case where the damper **120** is activated will be described. As shown in FIG. **6**, in a case where the rack bar **103** has reached the vicinity of the displacement limit (see a dashed arrow), such as a case where the wheel **112** of the self-propelled vehicle has been steered to the vicinity of the steering limit, the end portion of the integral displacement body **130** contacts the rack housing **104**, and the damper **120** is activated. In this case, the case where the damper **120** is activated includes a case where the end portion of the integral displacement body **130** has contacted the rack housing **104** with weak force or strong force.

First, in a case where the end portion of the integral displacement body **130** has contacted the rack housing **104** with weak force (low speed), the integral displacement body **130** slowly slidably displaces to the stud body **108** side relative to the socket main body **107**, as shown in FIG. **7** (see a dashed arrow). That is, the flow control valve **140** displaces to the buffer **123***a* side against the elastic force of the return elastic body **132** in the inner chamber **121**. In this case, in the first flow control valve **150**, the fluid **124** flows in from the large-diameter hole **157***a* side of the second flow hole **157** of the second flow body **156**, and flows toward a small-diameter hole **157***b* side.

However, in this case, since the flow control valve **140** slowly displaces in the inner chamber **121**, a force of pressing the second flow body **156** is smaller than the elastic force of the separation elastic body **159**. Thus, the second flow body **156** is not pressed against the first flow body **153** after having displaced to the first flow body **153** side. Consequently, in the first flow control valve **150**, the fluid **124** on the front side in the displacement direction flows, with slight flow resistance, to the rear side in the displacement direction through each of the second flow hole **157** of the second flow body **156** and the first flow hole **154** of the first flow body **153** (see a dashed arrow).

Since the second flow control valve **160** is the one-way valve blocking the flow of the fluid **124** from the front side to the rear side in the displacement direction of the flow control valve **140** when the integral displacement body **130** slidably displaces against the elastic force of the return elastic body **132**, the fluid **124** does not flow. Since the third flow control valve **170** is the valve allowing the fluid **124** to flow in both directions, i.e., the front side and the rear side, in the displacement direction of the flow control valve **140**, the fluid **124** flows, with slight flow resistance, from the front side to the rear side in the displacement direction of the flow control valve **140**.

Thus, the flow control valve **140** displaces to the buffer **123***a* side while generating ignorable extremely-small damping force. Accordingly, the integral displacement body **130** slowly slidably displaces to the stud body **108** side.

Thereafter, in a case where the rack bar **103** has displaced to a knuckle arm **111** side such that the end portion of the integral displacement body **130** is separated from the rack housing **104**, the integral displacement body **130** displaces to an original position by the elastic force of the return elastic body **132** (see FIG. **5**). In this case, in the first flow control valve **150**, the fluid **124** flows in from the first flow body **153** side, and flows toward a second flow body **156** side.

In this case, the second flow body **156** is positioned at an original position farthest from the first flow body **153** by the elastic force of the separation elastic body **159** and the pressing force of the fluid **124** having flowed from the first flow body **153**. Thus, in the first flow control valve **150**, the fluid **124** on the front side in the displacement direction flows, with slight flow resistance, to the rear side in the displacement direction through each of the first flow hole **154** of the first flow body **153** and the second flow hole **157** of the second flow body **156**.

Since the second flow control valve **160** is the one-way valve allowing the fluid **124** to flow from the front side to the rear side in the displacement direction of the flow control valve **140** when the integral displacement body **130** slidably displaces by the elastic force of the return elastic body **132**, the fluid **124** flows with slight flow resistance. Since the third flow control valve **170** is the valve allowing the fluid **124** to flow in both directions, i.e., the front side and the rear side, in the displacement direction of the flow control valve **140**, the fluid **124** flows, with slight flow resistance, from the front side to the rear side in the displacement direction of the flow control valve **140**.

Thus, the flow control valve **140** displaces to the buffer **123***b* side while generating ignorable extremely-small damping force. Accordingly, the integral displacement body **130** slidably displaces to the rack housing **104** side faster than the above-described displacement speed.

Next, in a case where the end portion of the integral displacement body **130** has contacted the rack housing **104** with strong force (high speed) (e.g., abrupt steering by the driver or collision of the wheel **112** with, e.g., a curbstone), the integral displacement body **130** rapidly slidably displaces to the stud body **108** side relative to the socket main body **107**. That is, the flow control valve **140** rapidly displaces to the buffer **123***a* side against the elastic force of the return elastic body **132** in the inner chamber **121**.

In this case, in the first flow control valve **150**, a force of pressing the second flow body **156** by the fluid **124** is greater than the elastic force of the separation elastic body **159**, and therefore, the second flow body **156** displaces to the first flow body **153** side and is pressed against the first flow body **153**, as shown in FIG. **8**. In this case, after the second flow body **156** has started displacing to the first flow body **153** side by action of the pressing force of the fluid **124** on the step portion **157***c* and the tapered portion **157***d* of the small-diameter hole **157***b*, the pressing force of the fluid **124** also acts on an end portion of the large-diameter hole **157***a*, and accordingly, the second flow body **156** displaces to the first flow body **153** side.

Moreover, in this case, since the first hole diameter restriction portion **155** and the second hole diameter restriction portion **158** are each formed at the positions facing the second flow hole **157** and the first flow hole **154**, the first hole diameter restriction portion **155** and the second hole diameter restriction portion **158** close the entirety of the second flow hole **157** and part of the first flow hole **154**. Thus, in the first flow control valve **150**, the fluid **124** does not flow (see a dashed arrow) upon displacement of the flow control valve **140** to the stud body **108** side. Further, in this case, the fluid **124** does not flow in the second flow control valve **160** as in the above-described case.

Since the fluid **124** can flow only in the third flow control valve **170** in the flow control valve **140**, extremely-great flow resistance is generated at the third flow control valve **170**. Thus, the flow control valve **140** displaces to the buffer **123***a* side against the extremely-great flow resistance by the third flow control valve **170**. Accordingly, the integral displacement body **130** slidably displaces to the stud body **108** side while generating extremely-great damping force. That is, the damper **120** can damp strong impact generated on the rack bar **103**.

Thereafter, in a case where the rack bar **103** has displaced to the knuckle arm **111** side such that the end portion of the integral displacement body **130** is separated from the rack housing **104**, the integral displacement body **130** displaces, as in the above-described case, to the original position by the elastic force of the return elastic body **132** (see FIG. **5**). That is, the flow control valve **140** displaces to the buffer **123***b* side while generating ignorable extremely-small damping force, and therefore, the integral displacement body **130** quickly slidably displaces to the rack housing **104** side.

The second flow body **156** is separated from the first flow body **153** by the elastic force of the separation elastic body **159** and the pressing force of the fluid **124** having flowed from the first flow body **153**, and returns to the original position. In this case, since the second hole diameter restriction portion **158** is formed at the position facing part of the first flow hole **154**, the second flow body **156** can guide part of the fluid **124** having flowed into the second flow body housing portion **151** through the first flow hole **154** to a second flow hole **157** side. In the first flow control valve **150**, the fluid **124** on the front side in the displacement direction flows, with slight flow resistance, to the rear side in the displacement direction through each of the first flow hole **154** of the first flow body **153** and the second flow hole **157** of the second flow body **156**, as described above.

As can be understood from description of the actuation method above, according to the first embodiment, in the damper **120**, the first flow body **153** and the second flow body **156** each having the first flow hole **154** and the second flow hole **157** causing the fluid **124** to flow are provided so as to elastically contact or separate from each other. Further, the first hole diameter restriction portion **155** is formed so as to close the entirety of the second flow hole **157**. With this configuration, in the damper **120** according to the above-described embodiment, in a case where great external force acts on between the first flow body **153** and the second flow body **156**, the first flow hole **154** and the second flow hole **157** are closed by the second hole diameter restriction portion **158** and the first hole diameter restriction portion **155**, and the flow of the fluid **124** is limited accordingly. Thus, the steering device **100** can be configured to damp the great external force and absorb great impact force.

Second Embodiment

Next, a second embodiment of a steering device including flow control valves and dampers according to the present invention will be described with reference to FIGS. **9** to **13**. A steering device **200** in the second embodiment is different from that of the first embodiment in that a damper **210** equivalent to the damper **120** in the first embodiment is not assembled with a socket main body **107**, but with a rack housing **104**. Thus, for the steering device **200** in the second embodiment, differences from the steering device **200** in the first embodiment will be mainly described, and description of common contents or corresponding contents between both embodiments will be omitted as necessary. In description of the second embodiment, the same reference numerals as those of the first embodiment will be used to represent components similar to those of the first embodiment.

(Configuration of Steering Device **200**)

In the steering device **200**, the cylindrical damper **210** is attached to a tip end portion of the rack housing **104** formed in a cylindrical shape. Further, a dust boot **201** is attached so as to cover the damper **210**. The dust boot **201** is a component for preventing contamination of the damper **210**. The dust boot **201** is formed in such a manner that an elastomer material such as a rubber material is formed in a cylindrical shape.

A rack bar **103** penetrating the rack housing **104** penetrates the cylindrical damper **210** attached to the tip end portion of the rack housing **104**. A rack end **106** is attached to a tip end portion of the rack bar **103**. In this case, an outer peripheral portion of the socket main body **107** attached to the rack bar **103** is formed so as to project in a flange shape from an outer peripheral portion of the rack bar **103**, and is formed so as to face an end portion of an integral displacement body **230**.

The damper **210** includes an inner chamber forming body **211**. The inner chamber forming body **211** is a component forming an inner chamber **217**, and is also a component for attaching the damper **210** to the rack housing **104**. The damper **210** is formed in such a manner that a metal material is formed in a cylindrical shape. That is, the inner chamber forming body **211** corresponds to the socket main body **107** in the first embodiment. One (the left side as viewed in the figure) end portion of an outer peripheral portion of the inner chamber forming body **211** is formed with an external thread portion **211***a* to be screwed into the rack housing **104**, and the other (the right side as viewed in the figure) end portion is formed with each of an oil supply port **212** and an accumulator housing portion **213**.

The oil supply port **212** is a flow path for injecting fluid **124** into the inner chamber **217** or discharging the fluid **124** from the inner chamber **217**. The oil supply port **212** is openably closed with a plug. The accumulator housing portion **213** and an accumulator **214** each correspond to the accumulator housing portion **135** and the accumulator **136** in the first embodiment. Wall forming bodies **215**, **216** are each screwed into both end portions of the inner chamber forming body **211**. The cylindrical inner chamber **217** is formed between these two wall forming bodies **215**, **216**.

The wall forming bodies **215**, **216** are components for forming both the right and left wall portions of the inner chamber **217** as viewed in the figure. The wall forming body **215**, **216** is formed in such a manner that a metal material is formed in a circular ring shape. That is, the wall forming body **215**, **216** corresponds to the wall forming body **122** in the first embodiment. Thus, the inner chamber **217** is formed in a circular-ring tubular shape extending in an axis direction among the wall forming bodies **215**, **216** and the later-described integral displacement body **230** inside the inner chamber forming body **211**. A return elastic body **218** is fitted onto an outer peripheral portion of the intermediate-coupling-body-**105**-side wall forming body **216** of these wall forming bodies **215**, **216**.

The return elastic body **218** is a component for elastically pressing a flow control valve **240** to the right end portion in the inner chamber **217** as viewed in the figure. The return elastic body **218** includes a metal coil spring. That is, the return elastic body **218** corresponds to the return elastic body **132** in the first embodiment. One (the left side as viewed in the figure) end portion of the return elastic body **218** elastically presses the wall forming body **216**, and the other (the right side as viewed in the figure) end portion elastically presses the integral displacement body **230** through a backing plate **218***a*.

The wall forming body **215**, **216** is provided with a buffer **221***a*, **221***b*, a sliding bush **222***a*, **222***b*, a seal ring **223***a*, **223***b*, and a dust seal **224***a*, **224***b* each corresponding to the buffer **123***a*, **123***b*, the sliding bush **125***a*, **125***b*, the seal ring **126***a*, **126***b*, and the dust seal **134** in the first embodiment.

The integral displacement body **230** is a component covering the inside of the inner chamber **217** in a radial direction and formed with the flow control valve **240**. The integral displacement body **230** is formed in such a manner that a metal material is formed in a cylindrical shape. That is, the integral displacement body **230** corresponds to the integral displacement body **130** in the first embodiment. The integral displacement body **230** is formed in such a cylindrical shape that the integral displacement body **230** is slidably fitted in inner peripheral surfaces of the wall forming bodies **215**, **216** through the sliding bushes **222***a*, **222***b*. In this case, the integral displacement body **230** is formed with such a length that the integral displacement body **230** projects from each end portion of the wall forming bodies **215**, **216**. The inner diameter of the integral displacement body **230** is formed with such a size that the rack bar **103** penetrates therethrough.

The backing plate **218***a* is attached in a fixed manner to one (the right side as viewed in the figure) end portion of an outer peripheral portion of the integral displacement body **230**, and each of a fixing sleeve **231** and the flow control valve **240** is attached to a portion from the other (the left side as viewed in the figure) end portion to a center portion in the axial direction. The fixing sleeve **231** is a component for pressing the flow control valve **240** fitted onto a small-diameter portion of the outer peripheral portion of the integral displacement body **230** against a large-diameter portion of the outer peripheral portion of the integral displacement body **230** and fixing the flow control valve **240** to the large-diameter portion. The fixing sleeve **231** is formed in such a manner that a metal material is formed in a cylindrical shape. The fixing sleeve **231** is assembled integrally with the integral displacement body **230** with fitted onto the outer peripheral portion of the integral displacement body **230**. The fixing sleeve **231** slides relative to the wall forming body **215** through the sliding bush **222***a*.

The flow control valve **240** is a tool causing the fluid **124** to flow with a limitation on the flow of the fluid **124** in the inner chamber **217** to control the flow of the fluid **124**, thereby generating damping force of the damper **210**. The flow control valve **240** corresponds to the flow control valve **140** in the first embodiment. The flow control valve **240** mainly includes each of a valve support **241**, a first flow control valve **150**, a second flow control valve **160**, and third flow control valves **170**.

The valve support **241** is a component formed with each of the first flow control valve **150**, the second flow control valve **160**, and the third flow control valves **170**. The valve support **241** is formed in such a manner that a metal material is formed in a flat-plate circular ring shape. That is, the valve support **241** is formed separately from the integral displacement body **230**, and is integrally attached to the integral displacement body **230** through the fixing sleeve **231**. A seal ring **242** corresponding to the seal ring **142** in the first embodiment is fitted onto an outer peripheral portion of the valve support **241**.

The first flow control valve **150**, the second flow control valve **160**, and the third flow control valve **170** are configured similar to those of the above-described embodiment, and therefore, description thereof will be omitted. The flow control valve **240** is attached to the outer peripheral portion of the integral displacement body **230** in such an orientation that a large-diameter portion **156***a* of a second flow body **156** of the first flow control valve **150** is opened on a buffer **221***a* side (the left side as viewed in the figure).

(Actuation of Steering Device **200**)

Next, actuation of the steering device **200** configured as described above will be described. As in the steering device **100** of the above-described embodiment, in the steering device **200**, each damper **210** is activated when the rack bar **103** has displaced to the vicinity of a right or left displacement limit with respect to a pinion gear **102***a*.

Specifically, the socket main body **107** does not contact the integral displacement body **230** within an area where the rack bar **103** does not reach the vicinity of the displacement limit, such as a case where a wheel **112** of a self-propelled vehicle is not steered to the vicinity of a steering limit, and therefore, the damper **210** is not activated (see FIG. **9**). In this case, in the damper **210**, the flow control valve **240** is pressed against the wall forming body **216** through the buffer

221*b* by elastic force of the return elastic body **218** in the inner chamber **217**. That is, the integral displacement body **230** is maintained in a state in which the integral displacement body **230** is elastically positioned at a position closest to a socket main body **107** side in the inner chamber **217**.

Further, the first flow control valve **150** is maintained in a state in which the second flow body **156** is positioned at a position farthest from the first flow body **153** by the elastic force of the separation elastic body **159**. That is, the first flow control valve **150** is in a state in which the flow of the fluid **124** is allowed.

Next, as shown in FIG. **11**, in a case where the rack bar **103** has reached the vicinity of the displacement limit, such as a case where the wheel **112** of the self-propelled vehicle has been steered to the vicinity of the steering limit, the socket main body **107** contacts the end portion of the integral displacement body **130**, and the damper **120** is activated.

First, in a case where the socket main body **107** has contacted the end portion of the integral displacement body **230** with weak force (low speed), the integral displacement body **230** slowly slidably displaces to a rack housing **104** side relative to the inner chamber forming body **211**, as shown in FIG. **12**. That is, the flow control valve **240** displaces to the buffer **221***a* side (the left side as viewed in the figure) against the elastic force of the return elastic body **218** in the inner chamber **217**. In this case, in the first flow control valve **150**, the fluid **124** flows in from a large-diameter hole **157***a* side of a second flow hole **157** of the second flow body **156**, and flows toward a small-diameter hole **157***b* side.

However, in this case, since the flow control valve **240** slowly displaces in the inner chamber **217**, a force of pressing the second flow body **156** is smaller than elastic force of a separation elastic body **159**. Thus, the second flow body **156** is not pressed against the first flow body **153** after having displaced to a first flow body **153** side. Consequently, in the first flow control valve **150**, the fluid **124** on the front side in the displacement direction flows, with slight flow resistance, to the rear side in the displacement direction through each of the second flow hole **157** of the second flow body **156** and a first flow hole **154** of the first flow body **153**.

Since the second flow control valve **160** is a one-way valve blocking the flow of the fluid **124** from the front side to the rear side in the displacement direction of the flow control valve **240** when the integral displacement body **230** slidably displaces against the elastic force of the return elastic body **218**, the fluid **124** does not flow. Since the third flow control valve **170** is a valve allowing the fluid **124** to flow in both directions, i.e., the front side and the rear side, in the displacement direction of the flow control valve **240**, the fluid **124** flows, with slight flow resistance, from the front side to the rear side in the displacement direction of the flow control valve **240**.

Thus, the flow control valve **240** displaces to the buffer **221***a* side while generating ignorable extremely-small damping force. Accordingly, the integral displacement body **230** slowly slidably displaces to the rack housing **104** side.

Thereafter, in a case where the rack bar **103** has displaced to a knuckle arm **111** side such that the socket main body **107** is separated from the end portion of the integral displacement body **230**, the integral displacement body **230** displaces to an original position by the elastic force of the return elastic body **218** (see FIG. **9**), as in the first embodiment. That is, the flow control valve **240** displaces to a buffer **221***b* side while generating ignorable extremely-small damping force. Accordingly, the integral displacement body **230** slidably displaces to the socket main body **107** side faster than the above-described displacement speed.

Next, in a case where the socket main body **107** has contacted the end portion of the integral displacement body **230** with strong force (high speed), the integral displacement body **230** rapidly slidably displaces to the rack housing **104** side relative to the inner chamber forming body **211**, as shown in FIG. **13**. That is, the flow control valve **240** rapidly displaces to the buffer **221***a* side against the elastic force of the return elastic body **218** in the inner chamber **217**.

In this case, in the first flow control valve **150**, a force of pressing the second flow body **156** by the fluid **124** is greater than the elastic force of the separation elastic body **159**, and therefore, the second flow body **156** displaces to the first flow body **153** side and is pressed against the first flow body **153**. In this case, since a first hole diameter restriction portion **155** and a second hole diameter restriction portion **158** are each formed at positions facing the second flow hole **157** and the first flow hole **154**, the first hole diameter restriction portion **155** and the second hole diameter restriction portion **158** close the entirety of the second flow hole **157** and part of the first flow hole **154**. Thus, in the first flow control valve **150**, the fluid **124** does not flow (see a dashed arrow) upon displacement of the flow control valve **240** to the rack housing **104** side. Further, in this case, the fluid **124** does not flow in the second flow control valve **160** as in the above-described case.

Since the fluid **124** can flow only in the third flow control valve **170** in the flow control valve **240**, extremely-great flow resistance is generated at the third flow control valve **170**. Thus, the flow control valve **240** displaces to the buffer **221***a* side against the extremely-great flow resistance by the third flow control valve **170**. Accordingly, the integral displacement body **230** slidably displaces to the rack housing **104** side while generating extremely-great damping force. That is, the damper **210** can damp strong impact generated on the rack bar **103**.

Thereafter, in a case where the rack bar **103** has displaced to the knuckle arm **111** side such that the socket main body **107** is separated from the end portion of the integral displacement body **230**, the integral displacement body **230** displaces, as in the above-described case, to the original position by the elastic force of the return elastic body **218** (see FIG. **9**). That is, the flow control valve **240** displaces to the buffer **221***b* side while generating ignorable extremely-small damping force, and therefore, the integral displacement body **230** quickly slidably displaces to the socket main body **107** side.

Implementation of the present invention is not limited to each of the above-described embodiments, and various changes can be made without departing from the object of the present invention. Note that in description of each variation, the same reference numerals are used to represent elements similar to those of the above-described embodiments and overlapping description thereof will be omitted.

For example, in each of the above-described embodiments, the first flow control valve **150** is configured such that the second hole diameter restriction portion **158** of the second flow body **156** closes part of the first flow hole **154** of the first flow body **153** and the first hole diameter restriction portion **155** of the first flow body **153** closes the entirety of the second flow hole **157** of the second flow body **156**. However, the first flow control valve **150** may only be required to be configured such that at least part of at least one of the first flow hole **154** or the second flow hole **157** is closed.

Thus, the first flow control valve **150** may be, for example, configured such that both the first flow hole **154** and the second flow hole **157** are fully closed, as shown in FIG. **14**. Alternatively, the first flow control valve **150** may be configured such that part of each of the first flow hole **154** and the second flow hole **157** is closed. In this case, the first flow control valve **150** may be, for example, configured such that part of the first flow hole **154** and part of the second flow hole **157** overlap with each other to ensure the flow of the fluid **124** when the second flow body **156** closely contacts the first flow body **153**, as shown in FIG. **15**. According to this configuration, the flow control valve **140**, **240** may be formed without the third flow control valve **170**.

The first flow control valve **150** may be configured such that one of the first flow hole **154** or the second flow hole **157** is fully closed and the other one of the first flow hole **154** or the second flow hole **157** is not closed at all. For example, as shown in FIG. **16**, in the first flow control valve **150**, the first hole diameter restriction portion **155** protruding in a columnar shape to the small-diameter hole **157***b* (the second flow hole **157**) side may be formed at a portion of the bottom portion of the second flow body housing portion **151** facing the small-diameter hole **157***b*. In this case, the second hole diameter restriction portion **158** of the second flow body **156** is omitted. According to this configuration, in the first flow control valve **150**, only the second flow hole **157** can be closed by contact of the second flow body **156** with the first hole diameter restriction portion **155**.

For example, as shown in each of FIGS. **17** and **18**, in the first flow control valve **150**, the second hole diameter restriction portion **158** protruding in a columnar shape toward a first flow hole **154** side is provided at a portion of the second flow body **156** facing the first flow hole **154** so that only the first flow hole **154** can be closed. In this case, the first hole diameter restriction portion **155** of the first flow body **153** is omitted. In these cases, the columnar first hole diameter restriction portion **155** and/or the columnar second hole diameter restriction portion **158** may be inserted into the small-diameter hole **157***b* (the second flow hole **157**) and/or the first flow hole **154** to close each hole.

In each of the above-described embodiments, the second flow hole **157** includes two holes, i.e., the large-diameter hole **157***a* and the small-diameter hole **157***b*. However, the second flow hole **157** may include one of the large-diameter hole **157***a* or the small-diameter hole **157***b*, or may include three or more holes having different inner diameters. At the second flow hole **157**, the tapered portion **157***d* is formed at an opening of the small-diameter hole **157***b* on the side opposite to the first flow body **153**. With this configuration, in the second flow body **156**, the fluid **124** can easily flow into the second flow hole **157**, and activation of the first flow control valve **150** can be easily stabilized. In the first flow control valve **150**, the fluid **124** easily flows into the second flow hole **157** so that a flow speed can be increased. Thus, the tapered portion **157***d* receives strong pressing force from the fluid **124**, and accordingly, the second flow body **156** can easily displace to the first flow body **153** side. However, the second flow hole **157** is not necessarily formed in the tapered shape, but may be formed in a straight shape, needless to say. Note that the first flow hole **154** may include multiple different holes or have a tapered shape at an opening.

In each of the above-described embodiments, the flow control valve **140**, **240** includes each of the second flow control valve **160** and the third flow control valves **170** in addition to the first flow control valve **150**. The second flow control valve **160** described herein can improve the displacement speed of the flow control valve **140**, **240** upon return displacement. The third flow control valve **170** can ensure the flow of the fluid **124** in a state in which the second flow body **156** closely contacts the first flow body **153** and the flow of the fluid **124** is fully blocked. However, the flow control valve **140**, **240** may be formed without at least one of the second flow control valve **160** or the third flow control valve **170**.

In each of the above-described embodiments, the damper **120**, **210** is configured such that the first flow control valve **150** is provided in the inner chamber **121**, **217**. However, the damper **120**, **210** may be configured such that the first flow control valve **150** is provided outside the inner chamber **121**, **217**. Note that the damper **120**, **210** may be configured such that the second flow control valve **160** and the third flow control valves **170** are also provided outside the inner chamber **121**, **217**.

In each of the above-described embodiments, the damper **120**, **210** includes the return elastic body **132**, **218**. However, the damper **120**, **210** may be formed without the return elastic body **132**, **218** in a case where the flow control valve **140**, **240** does not need to be constantly pressed to one side in the inner chamber **121**, **217**.

In the first embodiment, the socket main body **107** equivalent to the inner chamber forming body according to the present invention includes the solid bar body. However, the socket main body **107** can be formed according to a target to which the damper **120** is to be attached, and therefore, may be formed in a tubular shape, for example.

In the second embodiment, the integral displacement body **230** is formed in the tubular shape. However, the integral displacement body **230** can be formed according to a target to which the damper **210** is to be attached, and therefore, may be formed in a solid bar shape, for example.

In each of the above-described embodiments, the damper **120**, **210** is applied to the steering device **100**, **200**. However, the damper **120**, **210** may be, upon use thereof, attached to a device or a tool other than the steering device **100**, **200**, such as a suspension mechanism, a seat tilting mechanism, a door opening/closing mechanism, a mechanical device other than the self-propelled vehicle, an electrical device, a tool, or furniture.

LIST OF REFERENCE SIGNS

100 Steering Device
101 Steering Wheel
102 Steering Shaft
102*a* Pinion Gear
103 Rack Bar
103*a* Rack Gear
104 Rack Housing
105 Intermediate Coupling Body
106 Rack End
107 Socket Main Body (Inner Chamber Forming Body)
107*a* Ball Holding Portion
107*b* External Thread Portion
108 Stud Body
108*a* Ball Portion
110 Tie Rod
111 Knuckle Arm
112 Wheel
120 Damper
121 Inner Chamber
122 Wall Forming Body
123*a*, **123***b* Buffer
124 Fluid
125*a*, **125***b* Sliding Bush

126*a*, **126***b* Seal Ring
130 Integral Displacement Body
131 Elastic Body Holding Portion
132 Return Elastic Body
133 Dust Boot
134 Dust Seal
135 Accumulator Housing Portion
136 Accumulator
140 Flow Control Valve
141 Valve Support
142 Seal Ring
150 First Flow Control Valve
151 Second Flow Body Housing Portion
152 Retaining Ring
153 First Flow Body
154 First Flow Hole
155 First Hole Diameter Restriction Portion
156 Second Flow Body
156*a* Large-Diameter Portion
156*b* Small-Diameter Portion
157 Second Flow Hole
157*a* Large-Diameter Hole
157*b* Small-Diameter Hole
157*c* Step Portion
157*d* Tapered Portion
158 Second Hole Diameter Restriction Portion
159 Separation Elastic Body
160 Second Flow Control Valve
170 Third Flow Control Valve
200 Steering Device
201 Dust Boot
210 Damper
211 Inner Chamber Forming Body
211*a* External Thread Portion
212 Oil Supply Port
213 Accumulator Housing Portion
214 Accumulator
215, **216** Wall Forming Body
217 Inner Chamber
218 Return Elastic Body
218*a* Backing Plate
221*a*, **221***b* Buffer
222*a*, **222***b* Sliding Bush
223*a*, **223***b* Seal Ring
224*a*, **224***b* Dust Seal
230 Integral Displacement Body
231 Fixing Sleeve
240 Flow Control Valve
241 Valve Support
242 Seal Ring

What is claimed is:

1. A flow control valve provided at a flow path, in which fluid flows, to control a flow of the fluid by causing the fluid to flow with a limitation on the flow of the fluid, comprising:

a first flow body including a first flow hole through which the fluid flows;

a second flow body arranged so as to face the first flow body and including a second flow hole through which the fluid flows; and

a separation elastic body that produces an elastic force of separating the first flow body and the second flow body from a position at which the first flow body and the second flow body contact each other,

wherein at least one of the first flow body or the second flow body includes a hole diameter restriction portion that closes at least part of at least one of the second flow hole or the first flow hole when the first flow body and the second flow body contact each other.

2. The flow control valve according to claim **1**, wherein

the hole diameter restriction portion is provided only at one of the first flow body or the second flow body.

3. The flow control valve according to claim **1**, wherein

the hole diameter restriction portion is provided at each of the first flow body and the second flow body.

4. The flow control valve according to claim **1**, wherein

the hole diameter restriction portion is formed so as to fully close at least one of the second flow hole or the first flow hole.

5. The flow control valve according to claim **1**, further comprising:

a second flow body housing portion that movably houses the second flow body on a second flow body side with respect to the first flow body,

wherein the separation elastic body is provided between the first flow body and the second flow body in the second flow body housing portion.

6. The flow control valve according to claim **1**, wherein

the second flow body is, at an opening of the second flow hole on a side opposite to the first flow body, formed in such a tapered shape that a hole size decreases from an opening side toward a far side.

7. The flow control valve according to claim **1**, further comprising:

a one-way valve that causes the fluid to flow in a flow path different from the first flow body and the second flow body,

wherein the one-way valve allows the flow of the fluid from a first flow body side to a second flow body side, and blocks the flow of the fluid from the second flow body side to the first flow body side.

8. The flow control valve according to claim **1**, further comprising:

a flow restriction valve that causes the fluid to flow with a limitation on the flow of the fluid in a flow path different from the first flow body and the second flow body, and

the flow restriction valve causes the fluid to flow with the limitation on the flow of the fluid between the first flow body side and the second flow body side.

9. A damper including an inner chamber forming body forming an inner chamber housing fluid in a liquid-tight manner and damping external force received by the fluid by limiting a flow of the fluid, comprising:

the flow control valve according to claim **1**,

wherein the flow control valve causes the fluid to flow with the limitation on the flow of the fluid.

10. The damper according to claim **9**, further comprising:

a return elastic body that provides an elastic force of causing the fluid to flow from the first flow body side to the second flow body side in the flow control valve,

wherein the flow control valve is, in the inner chamber, provided displaceable relative to the inner chamber, and

the return elastic body provides the elastic force to one of the inner chamber forming body or the flow control valve to displace the one of the inner chamber forming body or the flow control valve relative to the other one of the inner chamber forming body or the flow control valve.

11. The damper according to claim **10**, further comprising:

an integral displacement body that displaces integrally with the flow control valve relative to the inner chamber,

wherein the inner chamber forming body is formed in a solid bar shape or a tubular shape,

the inner chamber is formed in a circular-ring tubular shape outside the inner chamber forming body,

the flow control valve is formed at a ring-shaped valve support to be fitted in the circular-ring tubular inner chamber, and

the integral displacement body is formed in a tubular shape slidably fitted onto the inner chamber forming body.

12. The damper according to claim **10**, further comprising:

an integral displacement body that displaces integrally with the flow control valve relative to the inner chamber,

wherein the inner chamber forming body is formed in a tubular shape,

the inner chamber is formed in a circular-ring tubular shape inside the inner chamber forming body,

the flow control valve is formed at a ring-shaped valve support to be fitted in the circular-ring tubular inner chamber, and

the integral displacement body is formed in a solid bar shape or a tubular shape to be slidably fitted in the inner chamber forming body.

13. A steering device including:

a steering shaft formed so as to extend in a bar shape and rotated by operation of a steering wheel,

a rack bar formed so as to extend in a rod shape and converting rotary motion of the steering shaft into reciprocating motion in an axis direction to transmit the reciprocating motion,

an intermediate coupling body coupled to each end portion of the rack bar to directly or indirectly couple a wheel targeted for steering to the each end portion,

a rack housing covering the rack bar, and

the damper according to claim **9**,

wherein the damper is provided between the rack housing and the rack bar or the intermediate coupling body to absorb impact from the wheel.

14. A steering device, including:

a steering shaft formed so as to extend in a bar shape and rotated by operation of a steering wheel,

a rack bar formed so as to extend in a rod shape and converting rotary motion of the steering shaft into reciprocating motion in an axis direction to transmit the reciprocating motion,

an intermediate coupling body coupled to each end portion of the rack bar to directly or indirectly couple a wheel targeted for steering to the each end portion,

a rack housing covering the rack bar, and

the damper according to claim **11**,

wherein the damper is provided between the rack housing and the rack bar or the intermediate coupling body to absorb impact from the wheel,

the inner chamber forming body is formed at the intermediate coupling body, and

the integral displacement body is formed at such a position that the integral displacement body contacts or separates from the rack housing by the reciprocating motion of the rack bar.

15. A steering device including:

a steering shaft formed so as to extend in a bar shape and rotated by operation of a steering wheel,

a rack bar formed so as to extend in a rod shape and converting rotary motion of the steering shaft into reciprocating motion in an axis direction to transmit the reciprocating motion,

an intermediate coupling body coupled to each end portion of the rack bar to directly or indirectly couple a wheel targeted for steering to the each end portion,

a rack housing covering the rack bar, and

the damper according to claim **12**,

wherein the damper is provided between the rack housing and the rack bar or the intermediate coupling body to absorb impact from the wheel,

the inner chamber forming body is formed at an end portion of the rack housing, and

the rack bar or the intermediate coupling body penetrates the integral displacement body, and the integral displacement body is formed at such a position that the rack bar or the intermediate coupling body contacts or separates from the integral displacement body by the reciprocating motion of the rack bar.

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